# Systemadministration

## Introduktion

Operativsystem

# Agenda

Operativsystem

1. Delar i ett operativsystem
2. Användargränssnitt
3. Drivrutiner
4. Processer
5. Historia
6. Säkerhet
7. Historia

# 1. Operating Systems

# Operating Systems

Operativsystem

En Liknelse…

- Säkerhetskontroll
- Flygledartorn
- Flygplan
- Flygbana

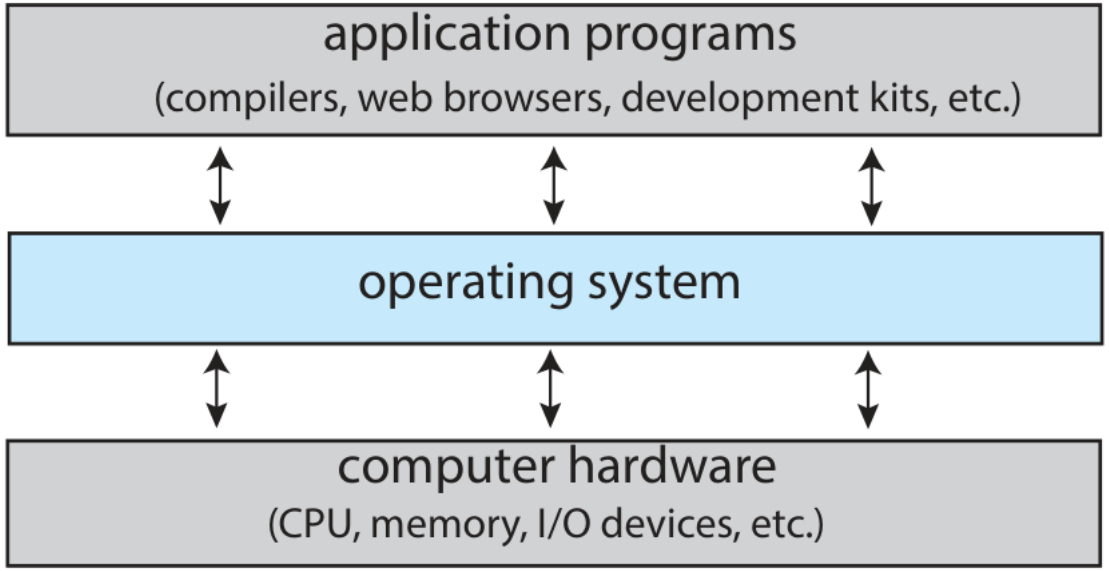
*Aerial view of airport with lots of airplanes during daytime*, by skyler_tv, 2016, Unsplash (https://unsplash.com/photos/9xdho4stJQ8). Unsplash license.

# Operating Systems

## Operativsystem

En Liknelse…

- Säkerhetskontroll
- Flygledartorn
- Flygplan
- Flygbana




From *Operating System Concepts* (10th edition), by A. Silberschatz, P. B. Galvin, G. Gagne, 2019, Wiley.
Used in accordance with Bonus Copyright Access.

# Operating Systems

Operativsystem

Består utav:

- **Kernel/kärna**
  Ett program som alltid körs
  ‘Är operativsystemet’
- **Systemprogram**
  Program som inte är en del av kärnan
- **Användarprogram**
  Övriga program, driftar inte systemet

# Operating Systems

Operativsystem

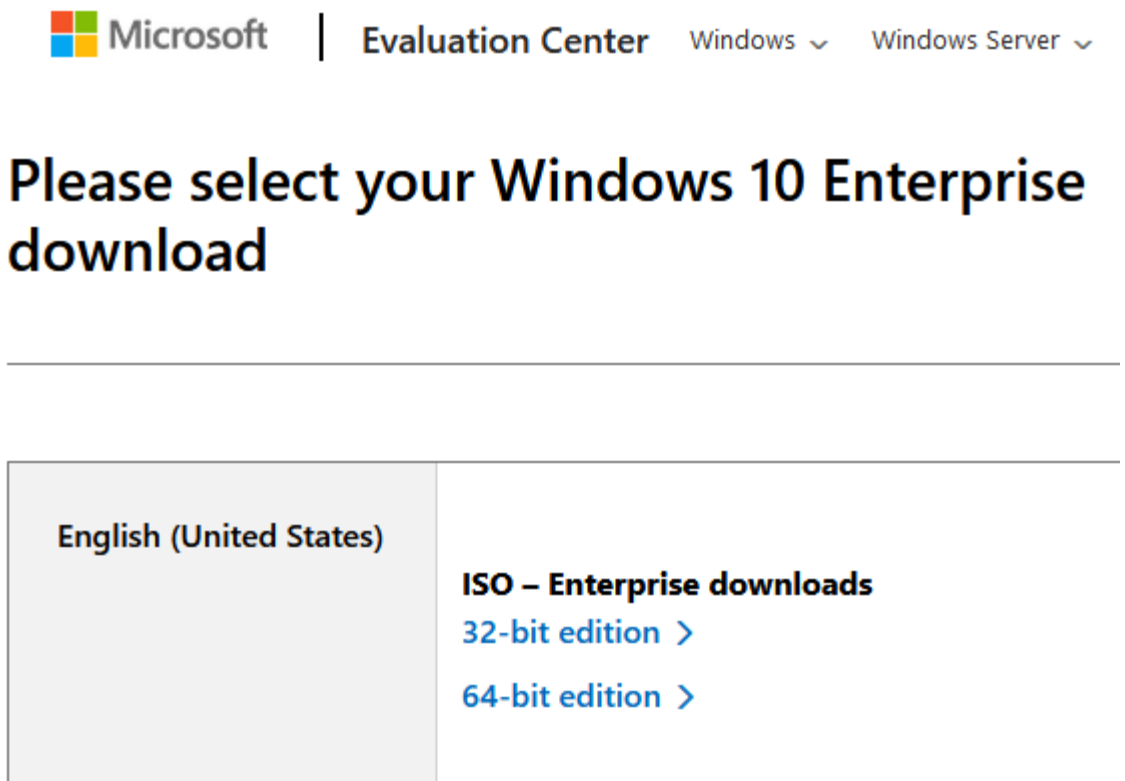
Tillhandahåller funktioner för bl.a:

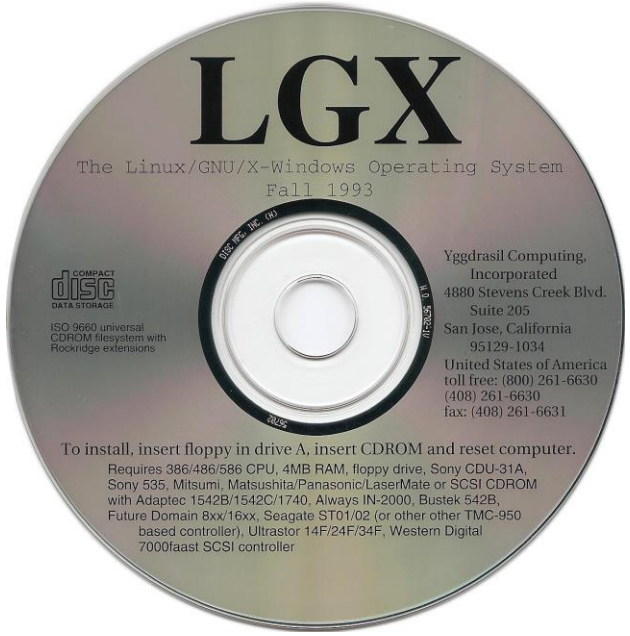
- Ett användargränssnitt
- Exekvering av program
- Filsystem
- Kommunikation mellan processer
- Tillgång till hårdvara
- Säkerhet

# 2. Operativsystems installation

# Hur installerar man ett OS?

1. Ladda ner en **installation image**
   - **ISO-fil ’skivavbildning’**



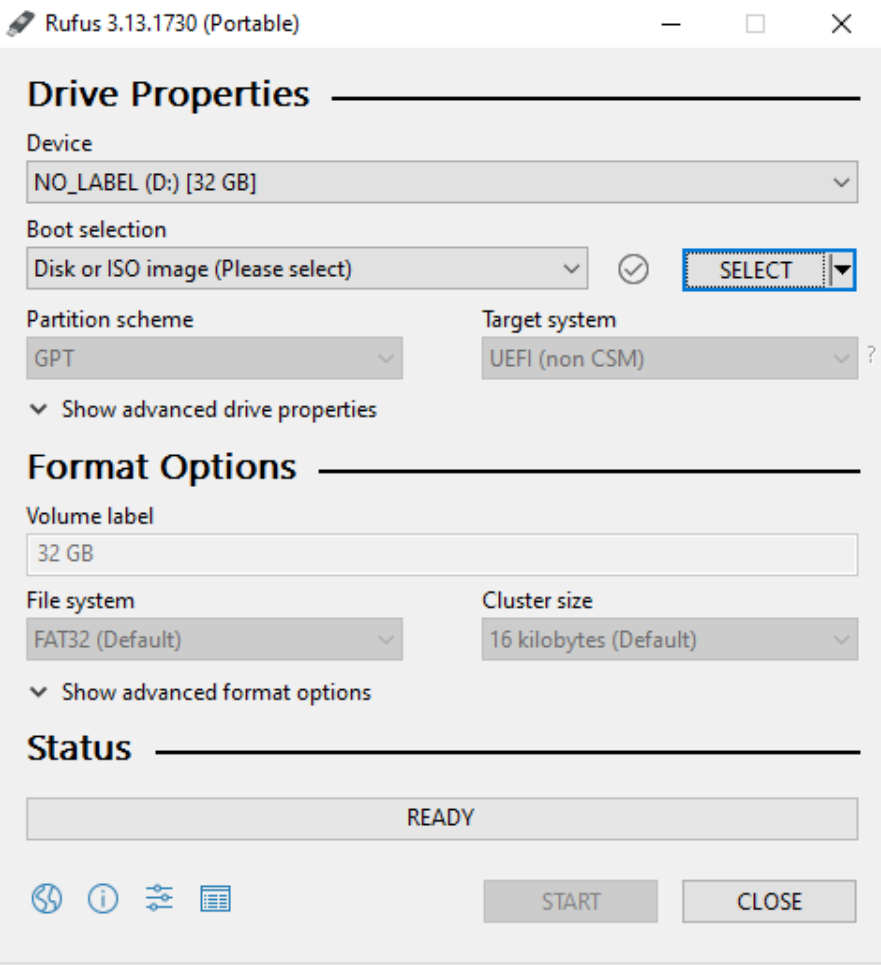




*CD-ROM of the LGX Yggdrasil Linux Distribution release “Fall 1993”*, by Ralfk, 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Lgx_yggdrasil_fall_1993.jpg). CC BY-SA 3.0.

# Hur installerar man ett OS?

2. Skapa ett installationsmedia
   - CD/DVD-skiva eller USB-minne
   - Program för att skapa installationsmedium

     *Rufus*
     *balenaEtcher*
     *Microsoft Media Creation Tool*



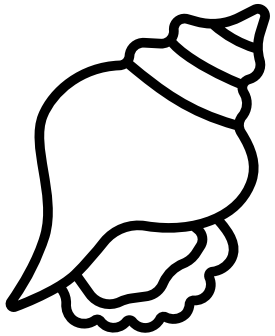
# Hur installerar man ett OS?

3. Montera installationsmediet och starta datorn

# 2. Användargränssnitt

# **Shell**

Skal



“[…] is the part of the operating system that interfaces with the outside world”

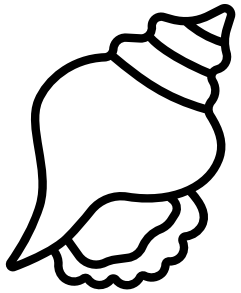
Raymond, E. S. (2003). shell. *The on-line hacker Jargon File, version 4.4.7*. Retrived from http://www.catb.org/jargon/html/S/shell.html

Det **användargränssnitt** som människor använder

**User Interface (UI)**

# User Interface

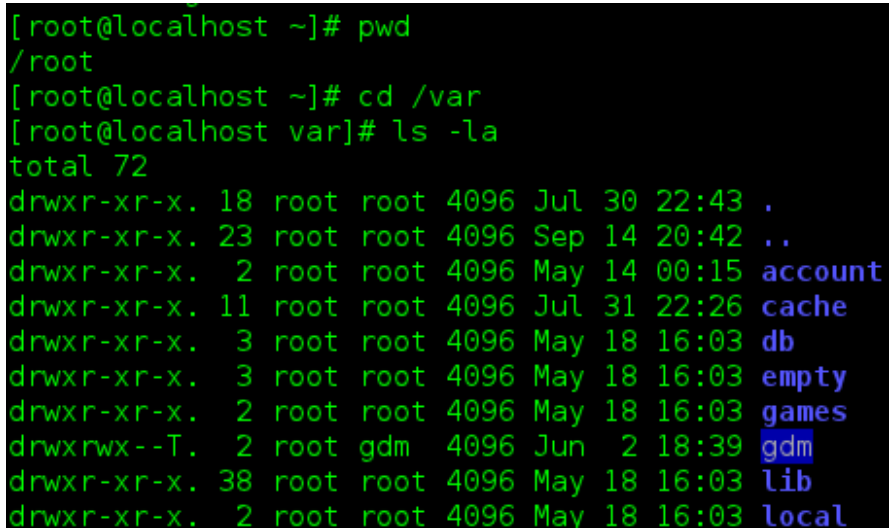
## Användargränssnitt



**GUI** **G**raphical
**U**ser
**I**nterface

**CLI** **C**ommand-**L**ine
**I**nterface

Microsoft Windows 10 Home, version 22H2, by Microsoft Coporation, 2022, Wikimedia Commons (https://en.wikipedia.org/w/index.php?curid=71854645). Fair use/I enlighet med citaträtt, trademark of Microsoft Corporation.



```
[root@localhost ~]# pwd
/root
[root@localhost ~]# cd /var
[root@localhost var]# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
drwxr-xr-x.  2 root root 4096 May 14 00:15 account
drwxr-xr-x. 11 root root 4096 Jul 31 22:26 cache
drwxr-xr-x.  3 root root 4096 May 18 16:03 db
drwxr-xr-x.  3 root root 4096 May 18 16:03 empty
drwxr-xr-x.  2 root root 4096 May 18 16:03 games
drwxrwx--T.  2 root gdm  4096 Jun  2 18:39 gdm
drwxr-xr-x. 38 root root 4096 May 18 16:03 lib
drwxr-xr-x.  2 root root 4096 May 18 16:03 local
```

*Sample of Bash in GNOME Terminal. Screenshot taken in Fedora. Results of ping, pwd, cd, yum and ls command.*, by ZxxZxxZ, 2011, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=16485488). No license.

# Application UI

Operativsystem kan vara GUI- eller CLI-baserade

Även program kan vara GUI- eller CLI-baserade

Ex.

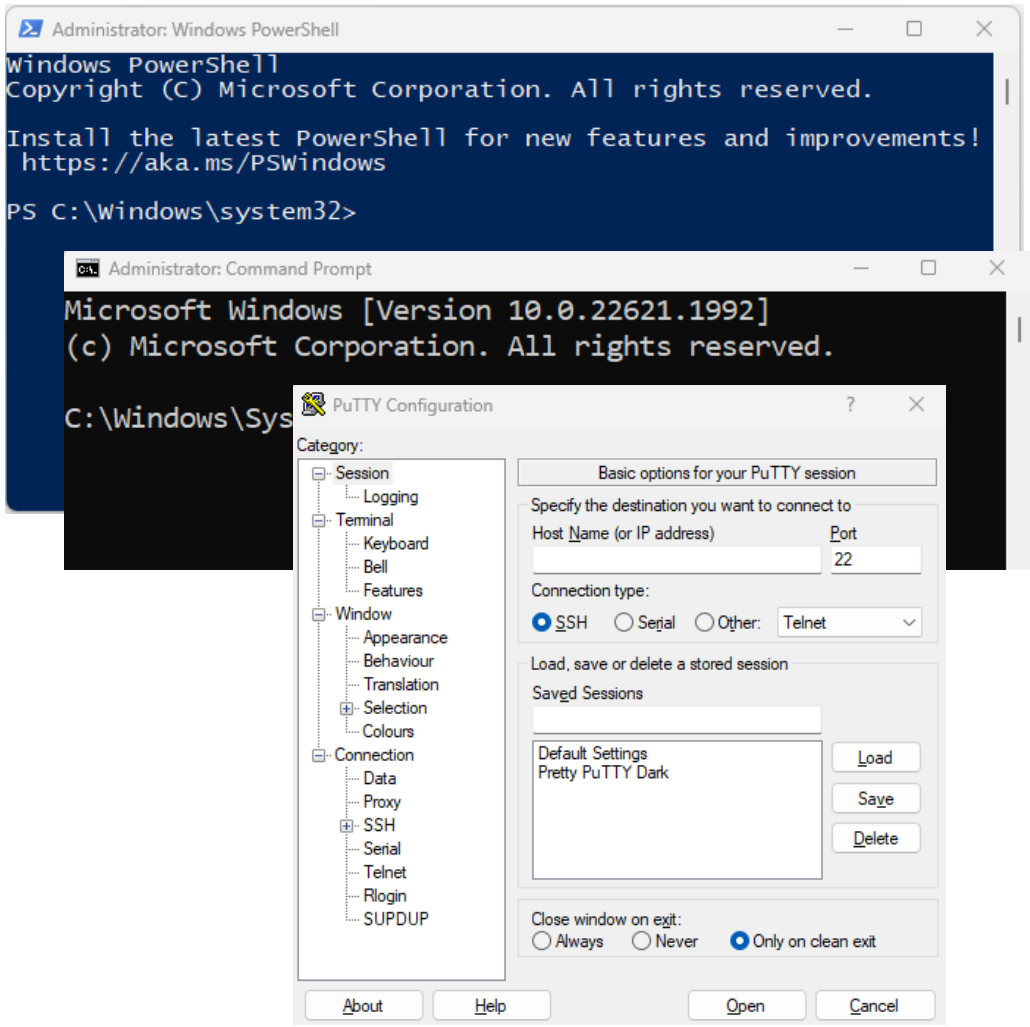
*This screenshot shows the X Window System running Tom's Window Manager (twm), and a number of client apps: the xlogo, an xterm, oclock, xbiff, xman, and xload. This screenshot is reminiscent of a typical Unix graphical desktop from the late 1980s to early 1990s.*, by Liberal Classic, 2007, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=1680280). MIT license.

# ’CLI-programs’

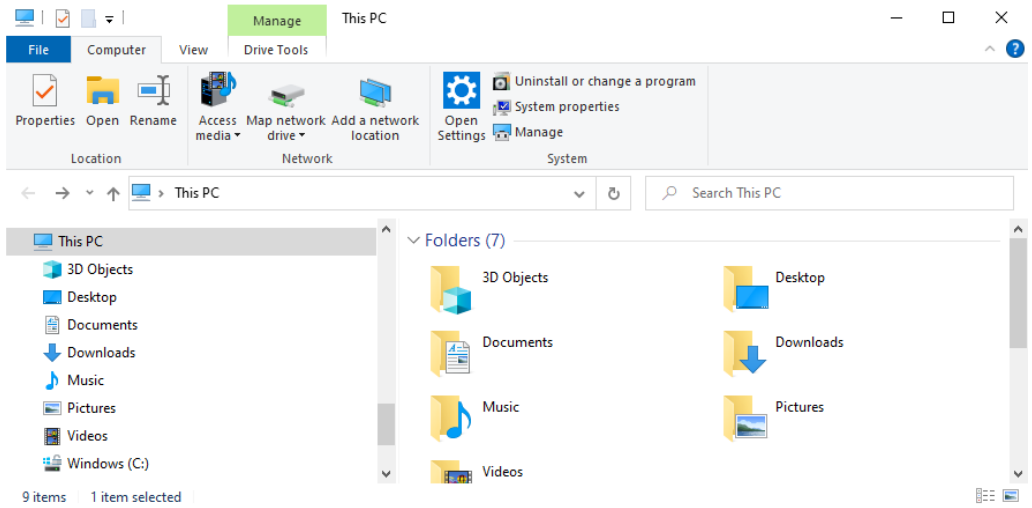
T.ex. cmd & PuTTY

Samlingsnamn b.la.:

- **Command-line interpreter**
- **Terminal emulator**
- Virtual teletype (**TTY**)
- Pseudoterminal (**PTS**)



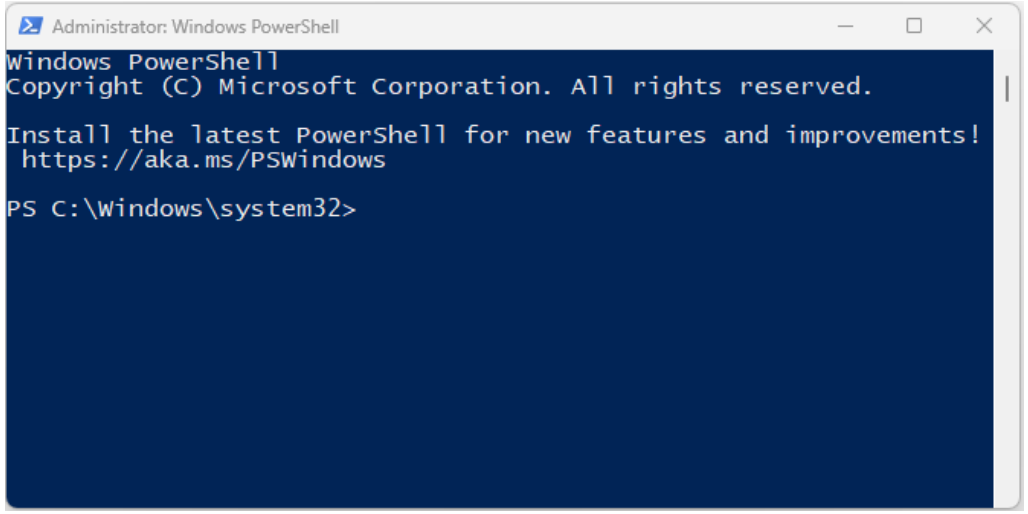
# GUI




Enklare att komma igång

Kan gissa sig fram

# CLI

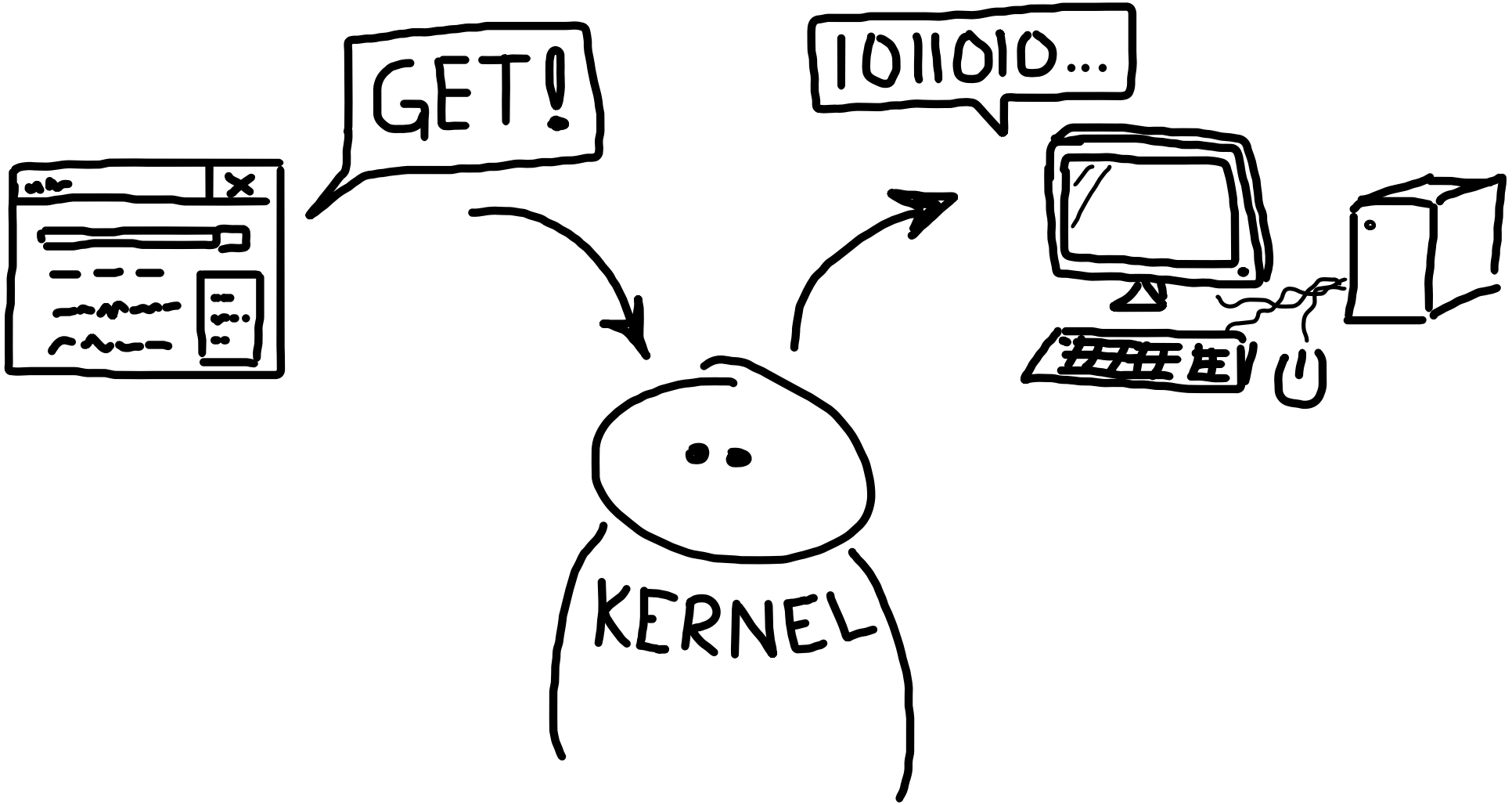


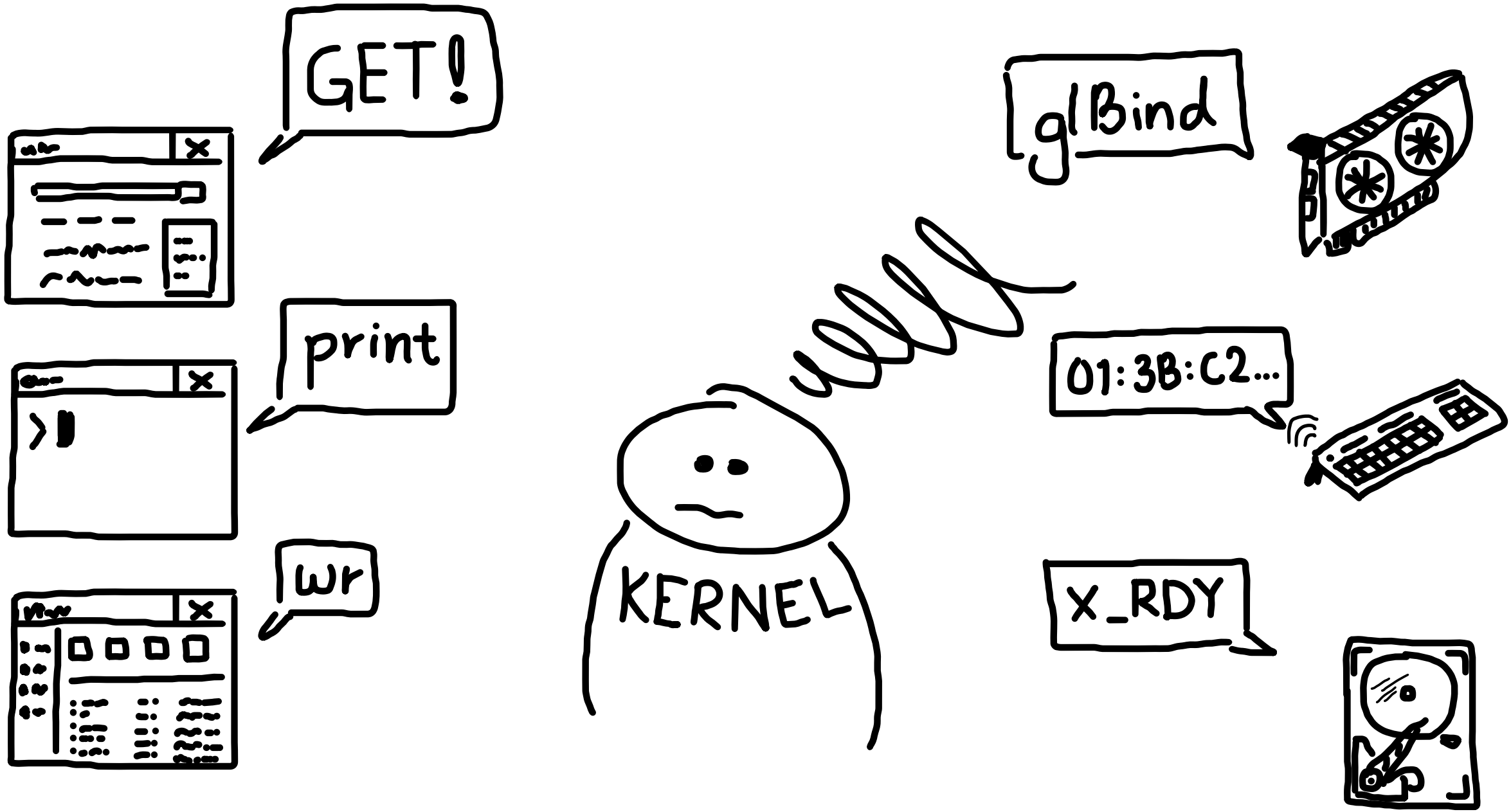

Ofta snabbare

Kan automatiseras

# 3. Drivrutiner

# Operativsystemet agerar mellanhand mellan mjukvara och hårdvara

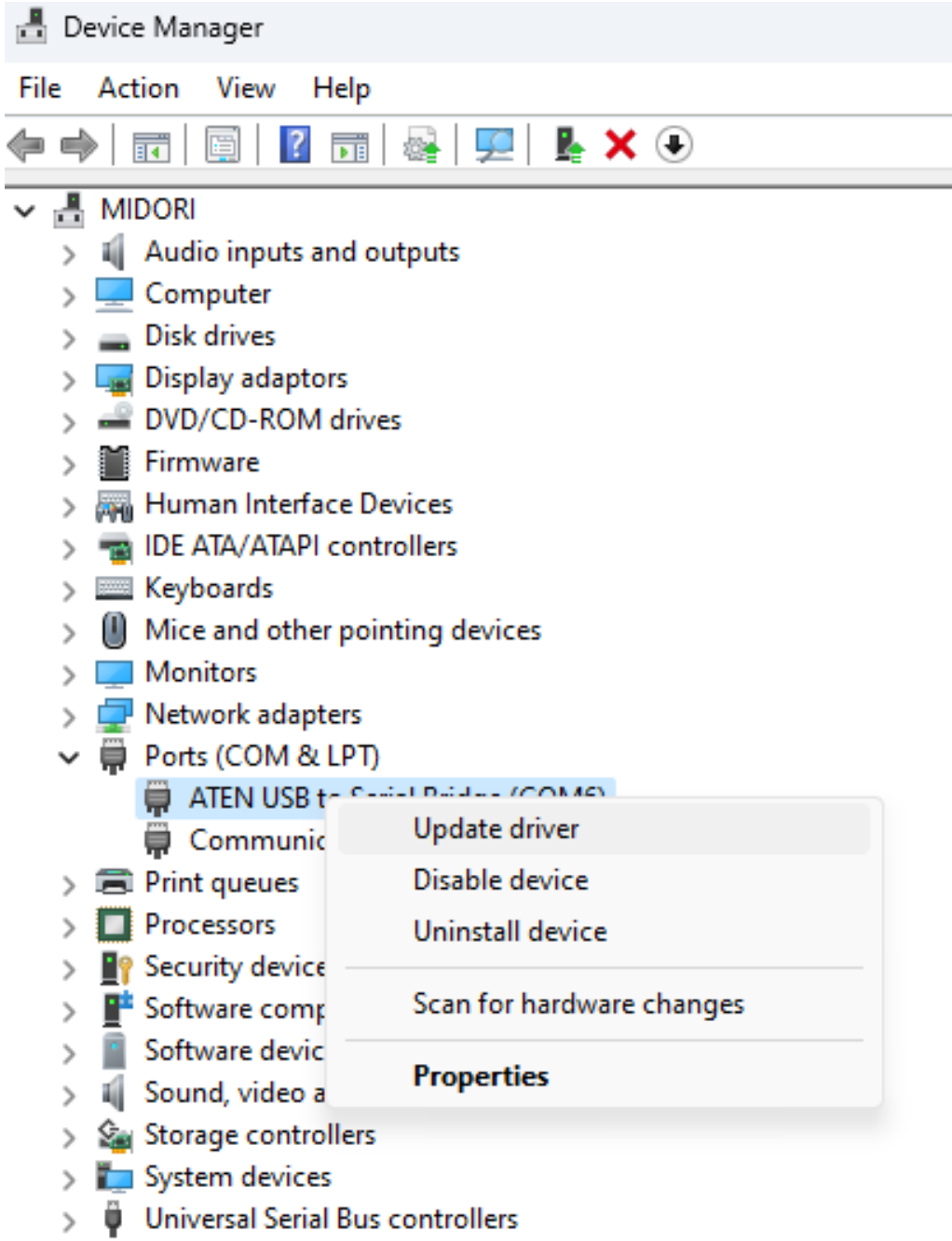



GET!
print
wr
KERNEL
glBind
01:3B:C2...
X_RDY

# Device Drivers

Drivrutiner

*Drivrutiner* är mjukvara som beskriver hur operativsystemet ska kommunicera med en enhet (eng. *device*)

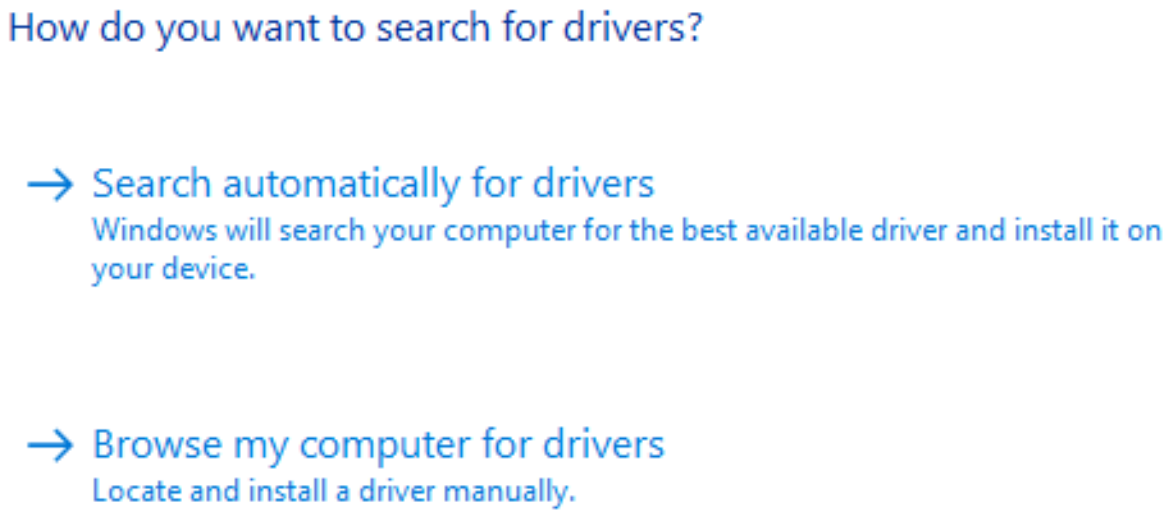


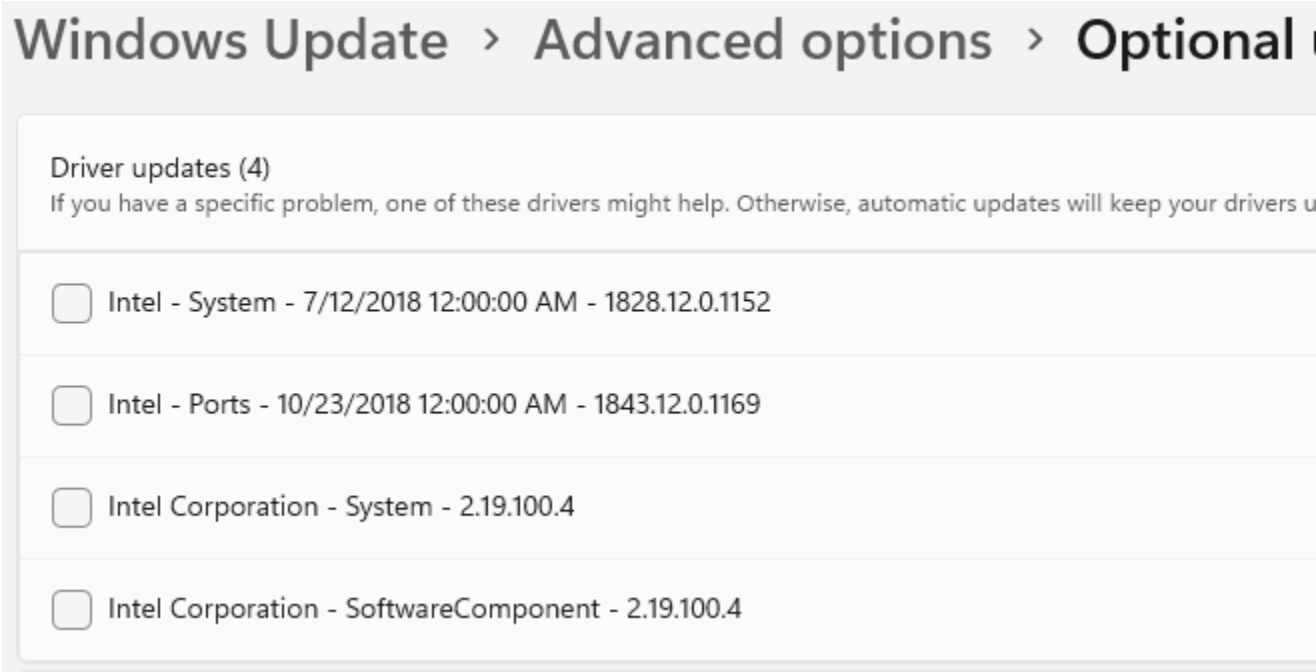
# Device Drivers

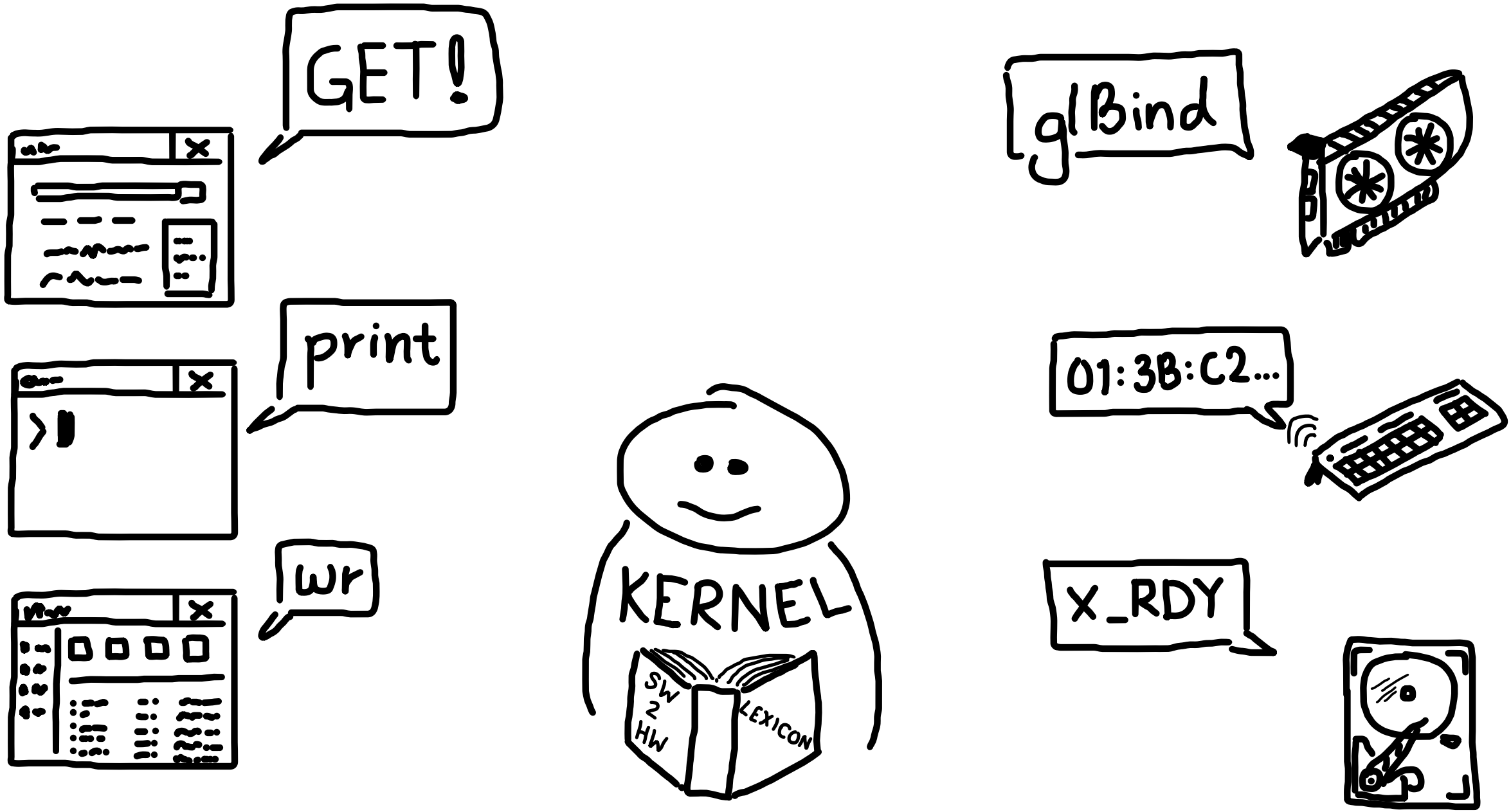
## Drivrutiner

I Windows har det blivit allt enklare att hitta rätt *drivrutiner*

Installeras ofta automatiskt i bakgrunden




GET!
print
wr
KERNEL
SW 2 HW
LEXICON
glBind
01:3B:C2...
X_RDY

# 4. Processer

# Process

Ett program är ett uppsättning instruktioner

“[…] a program in execution is termed a **process**”

Silberscahtz, A., Galvin, P. B. & Gagne, G. (2019). Multiprogramming and Multitasking. *Operating System Concepts 9th ed* (pp. 23). Wiley.

Ett program som körs = process

# Process Metadata

Håller operativsystemet bland annat koll på:

| | |
|---|---|
| **PID** (Process Id) | Unikt nummer, identifierar processen |
| **User** | Vilken användaren som kör processen |
| **PPID** (Parent PID) | Vilken överordnad process som hanterar processen (om någon) |

| Process | PID | User Name | Command Line |
|---|---|---|---|
| csrss.exe | 129820 | NT AUTHORITY\SYSTEM | |
| winlogon.exe | 136464 | NT AUTHORITY\SYSTEM | C:\Windows\System32\WinLogon.exe -SpecialSession |
| fontdrvhost.exe | 150180 | Font Driver Host\UMFD-7 | "fontdrvhost.exe" |
| dwm.exe | 151184 | Window Manager\DWM-7 | "dwm.exe" |
| igfxEM.exe | 133976 | MIDORI\Viktor | "C:\Windows\System32\DriverStore\FileRepository\igdlh64.inf_amd64_3faa07c7a01ca5fb\igfxEM.exe" |
| explorer.exe | 150176 | MIDORI\Viktor | C:\Windows\Explorer.EXE |
| OUTLOOK.EXE | 118540 | MIDORI\Viktor | "C:\Program Files\Microsoft Office\Root\Office16\OUTLOOK.EXE" |
| ai.exe | 156796 | MIDORI\Viktor | "C:\Program Files\Microsoft Office\root\vfs\ProgramFilesCommonX64\Microsoft Shared\Office16\ai.ex... |
| msedgewebview2.exe | 100228 | MIDORI\Viktor | "C:\Program Files (x86)\Microsoft\EdgeWebView\Application\115.0.1901.203\msedgewebview2.exe" -... |
| msedgewebview2.exe | 148288 | MIDORI\Viktor | "C:\Program Files (x86)\Microsoft\EdgeWebView\Application\115.0.1901.203\msedgewebview2.exe" -... |
| msedgewebview2.exe | 103028 | MIDORI\Viktor | "C:\Program Files (x86)\Microsoft\EdgeWebView\Application\115.0.1901.203\msedgewebview2.exe" -... |

# Process Types

En process kan starta andra processer

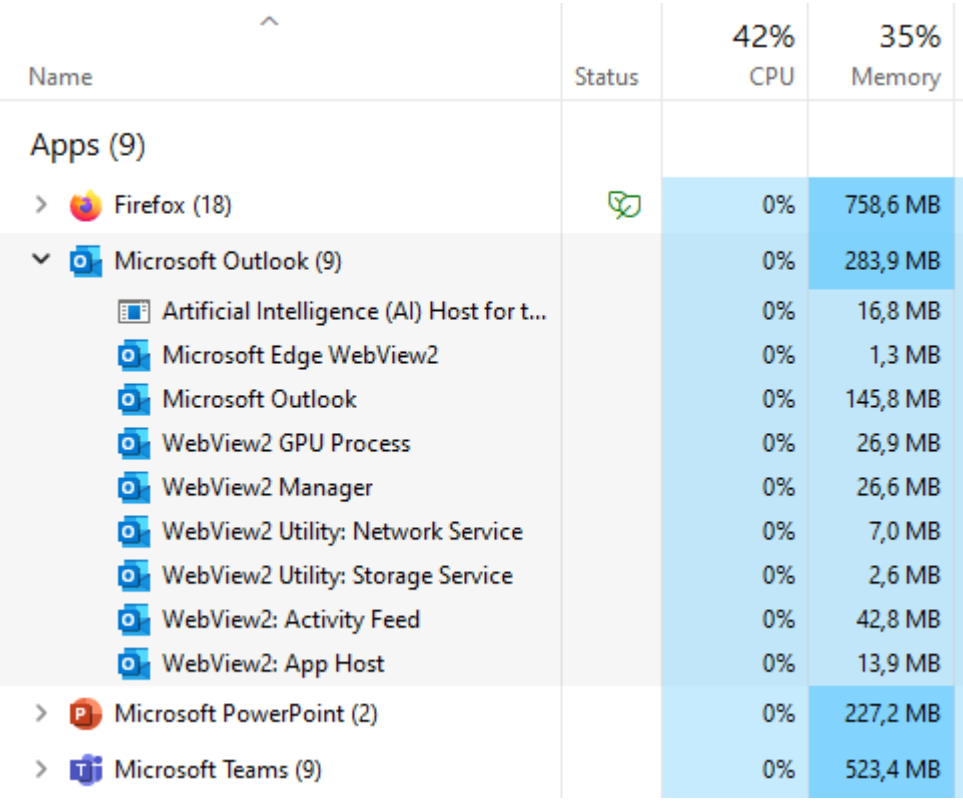
**Parent** **Child**

| Process | CPU | Private Bytes | Working Set | PID | Description |
|---|---|---|---|---|---|
| OUTLOOK.EXE | < 0.01 | 205 092 K | 355 692 K | 118540 | Microsoft Outlook |
| ai.exe | < 0.01 | 19 368 K | 32 384 K | 156796 | Artificial Intelligence (AI) Host for the Microsoft® Windo |
| msedgewebview2.exe | < 0.01 | 34 152 K | 101 296 K | 100228 | Microsoft Edge WebView2 |
| msedgewebview2.exe | | 2 092 K | 8 140 K | 148288 | Microsoft Edge WebView2 |
| msedgewebview2.exe | | 44 448 K | 66 196 K | 103028 | Microsoft Edge WebView2 |
| msedgewebview2.exe | < 0.01 | 10 704 K | 35 100 K | 40252 | Microsoft Edge WebView2 |
| msedgewebview2.exe | | 6 000 K | 17 500 K | 85380 | Microsoft Edge WebView2 |
| msedgewebview2.exe | | 55 136 K | 97 420 K | 138968 | Microsoft Edge WebView2 |
| msedgewebview2.exe | | 23 520 K | 61 052 K | 120952 | Microsoft Edge WebView2 |
| SecurityHealthSystray.exe | | 1 892 K | 10 716 K | 152812 | Windows Security notification icon |
| RAVCpl64.exe | | 4 172 K | 15 960 K | 47664 | Realtek HD Audio Manager |
| OneDrive.exe | 0.54 | 46 252 K | 142 664 K | 72388 | Microsoft OneDrive |

# Flera processer

För att göra saker parallellt kan flera processer användas

Kan ses som flera program som kommunicerar sinsemellan



| Name | Status | 42% CPU | 35% Memory |
|---|---|---|---|
| Apps (9) | | | |
| Firefox (18) | | 0% | 758,6 MB |
| Microsoft Outlook (9) | | 0% | 283,9 MB |
| Artificial Intelligence (AI) Host for t... | | 0% | 16,8 MB |
| Microsoft Edge WebView2 | | 0% | 1,3 MB |
| Microsoft Outlook | | 0% | 145,8 MB |
| WebView2 GPU Process | | 0% | 26,9 MB |
| WebView2 Manager | | 0% | 26,6 MB |
| WebView2 Utility: Network Service | | 0% | 7,0 MB |
| WebView2 Utility: Storage Service | | 0% | 2,6 MB |
| WebView2: Activity Feed | | 0% | 42,8 MB |
| WebView2: App Host | | 0% | 13,9 MB |
| Microsoft PowerPoint (2) | | 0% | 227,2 MB |
| Microsoft Teams (9) | | 0% | 523,4 MB |

# Threads

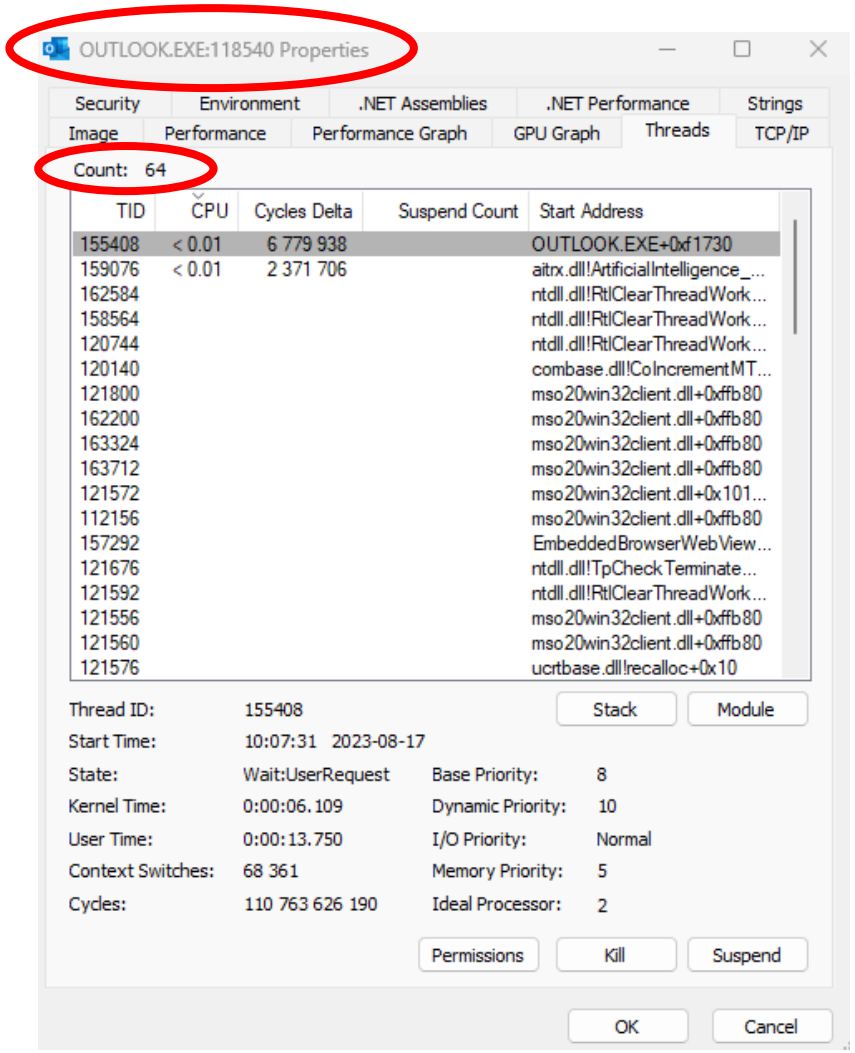
Trådar

Obs! Inte samma sak som **trådar** i *Hyper Threading* (HT) eller *Simultaneous Multi-Threading* (SMT)

En process består av minst en tråd

Varje tråd kan köras nästan oberoende av varandra

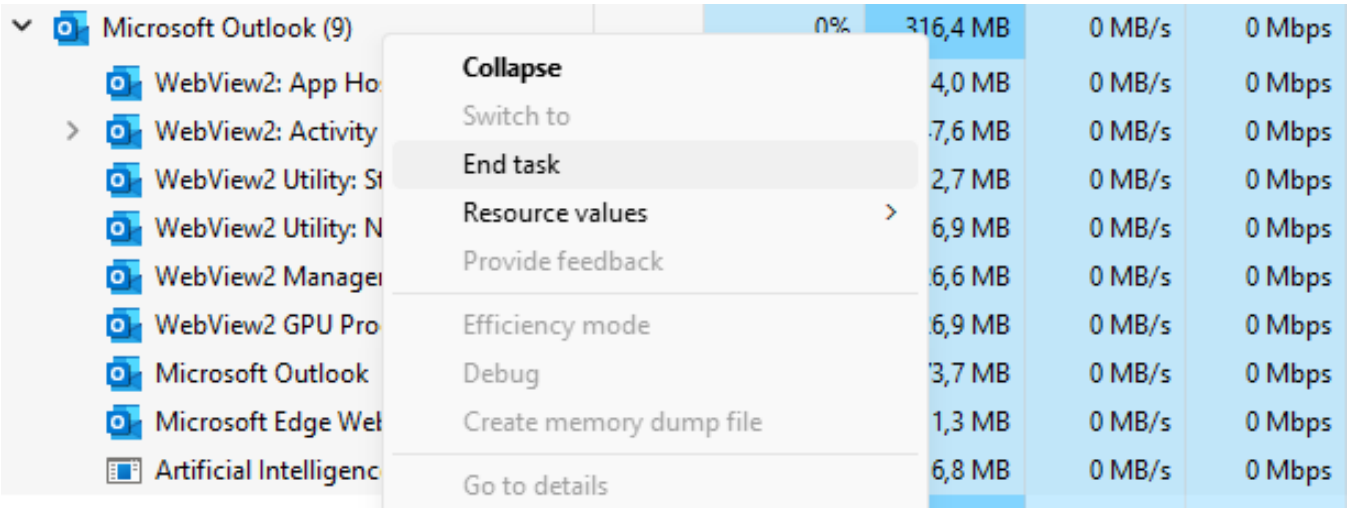
Om en process har flera trådar kan den utföra flera saker parallellt



# Signals

Operativsystemet och processer använder så kallade signaler för att kommunicerar

T.ex. End task

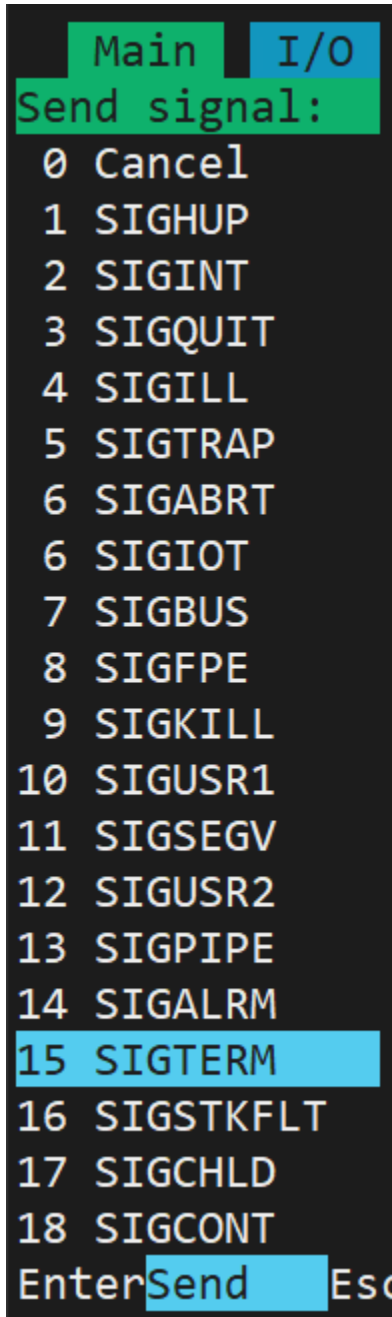


# Signals

UNIX-liknande operativsystem har många fler…

De viktigaste:

| Signal | Name | Description |
|---|---|---|
| **HUP** | Hangup | Usually restarts the process |
| **TERM** | Terminate | Request termination, the process cleans up and exits |
| **KILL** | Kill | The kernel forces immediate termination without signaling the process |



# Process States

En process kan vara…

igång

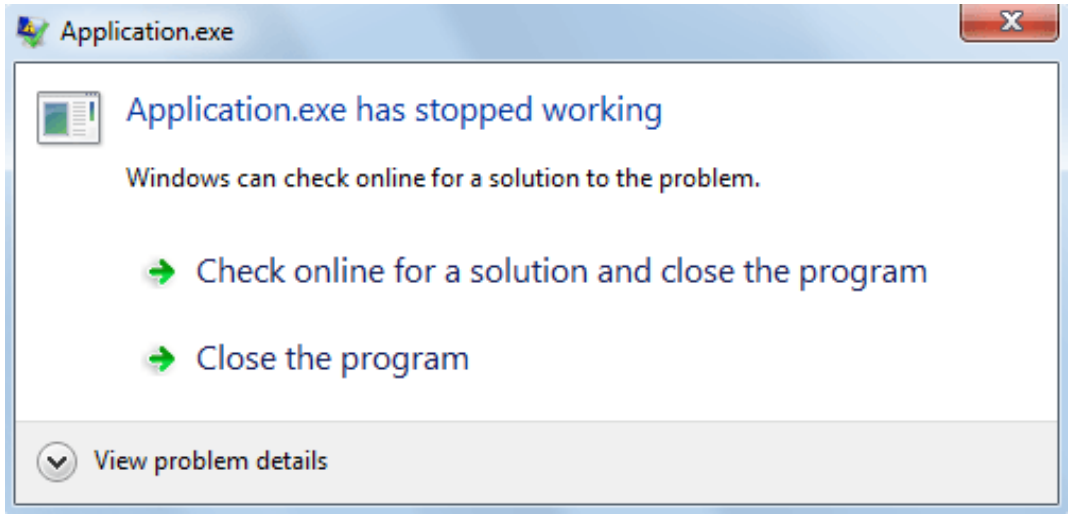
- körs, running

avstängd

- terminated, exited, finished, stopped, ’crashed’

’hängt sig’

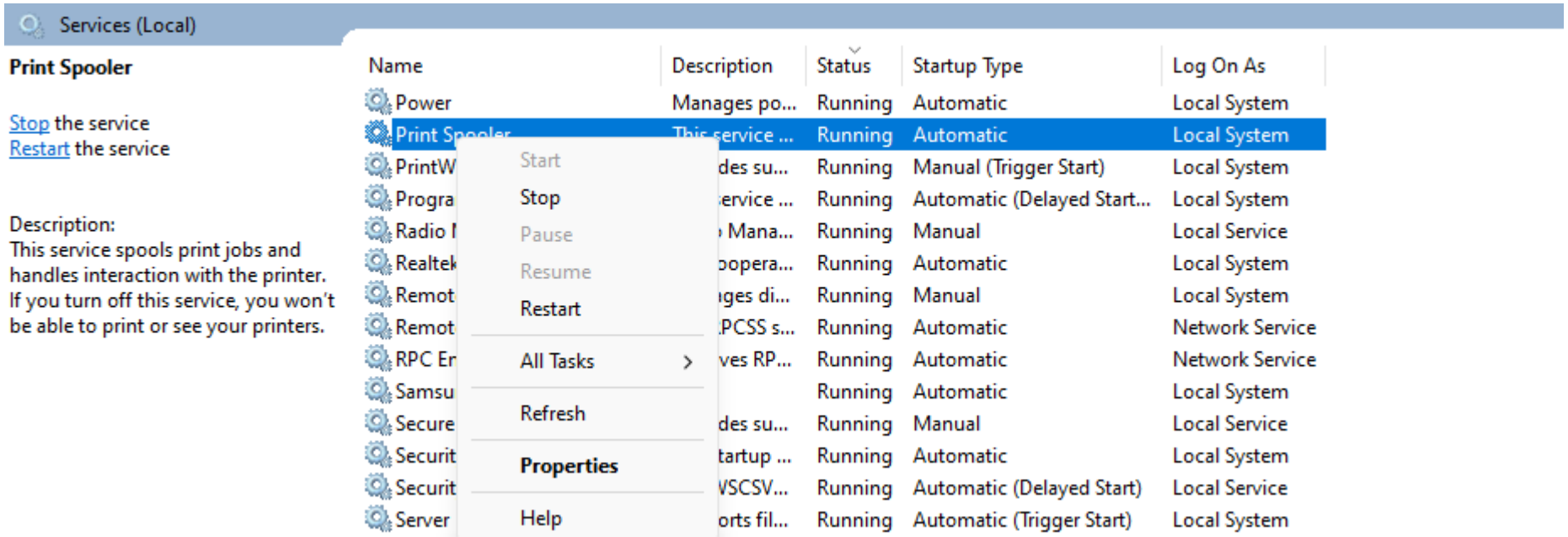
- unresponsive, ’frozen’, ’hung’/’hangs’



# Bakgrundsprocesser

Processer utan användargränssnitt

- Windows: *Services*
- Andra OS: *Daemons* (sv. *Demoner*)



# 7. Historia

# Operativsystems utveckling*

1950-talet

Program skrevs för en specifik dator

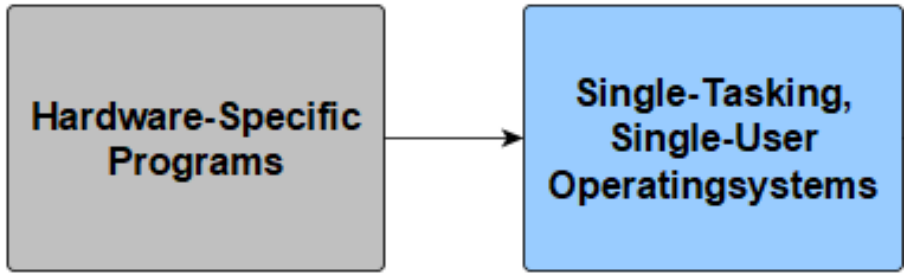
Finns inget operativsystem

Ett program kan köras i taget

*Gravt förenklad

# Operativsystems utveckling*

först 1956




Enkla operativsystem börjar växa fram

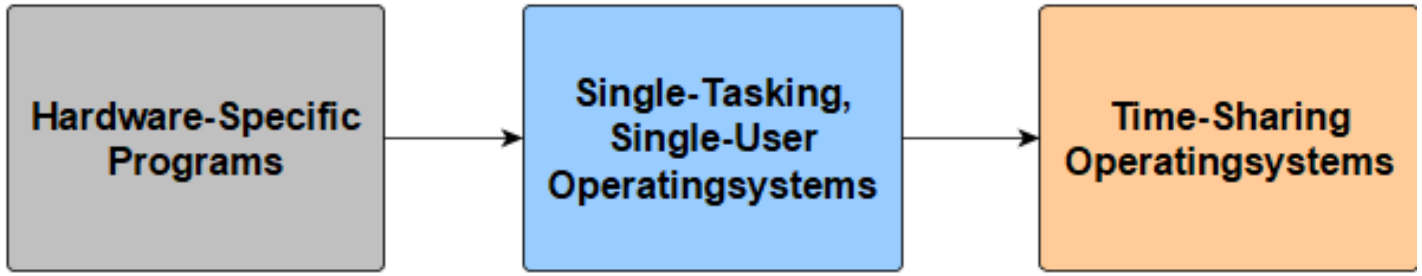
Underlättar programutveckling, behöver inte ta samma hänsyn till hårdvaran

Ett program kan köras i taget

*Gravt förenklad

# Operativsystems utveckling*

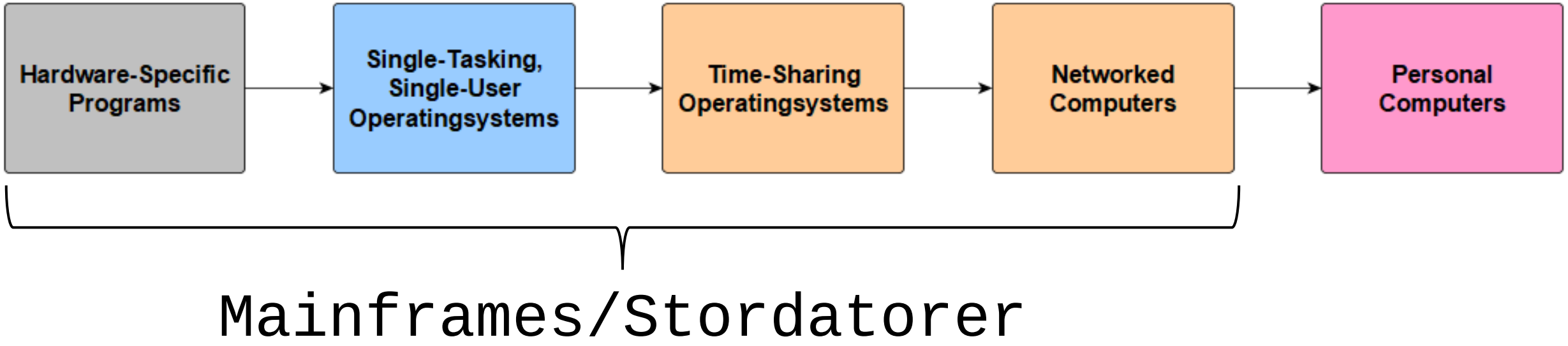
först 1961




Möjliggör att flera användare och flera program kan använda samma dator parallellt

*Gravt förenklad

# Operativsystems utveckling*

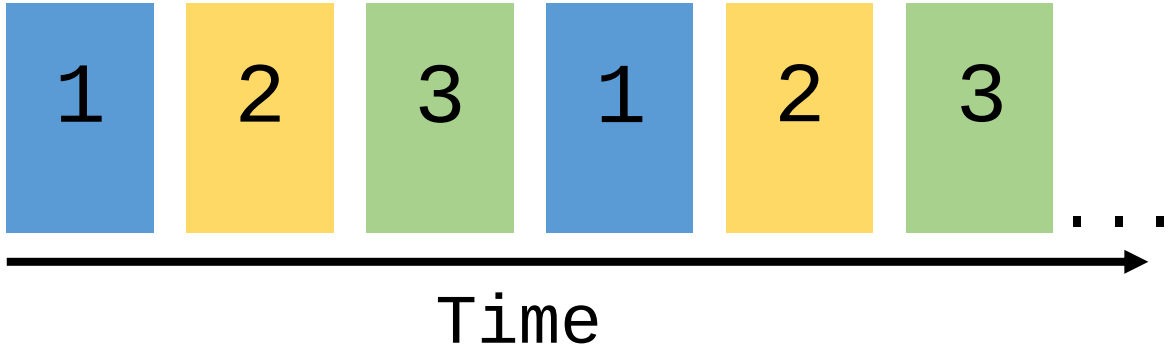


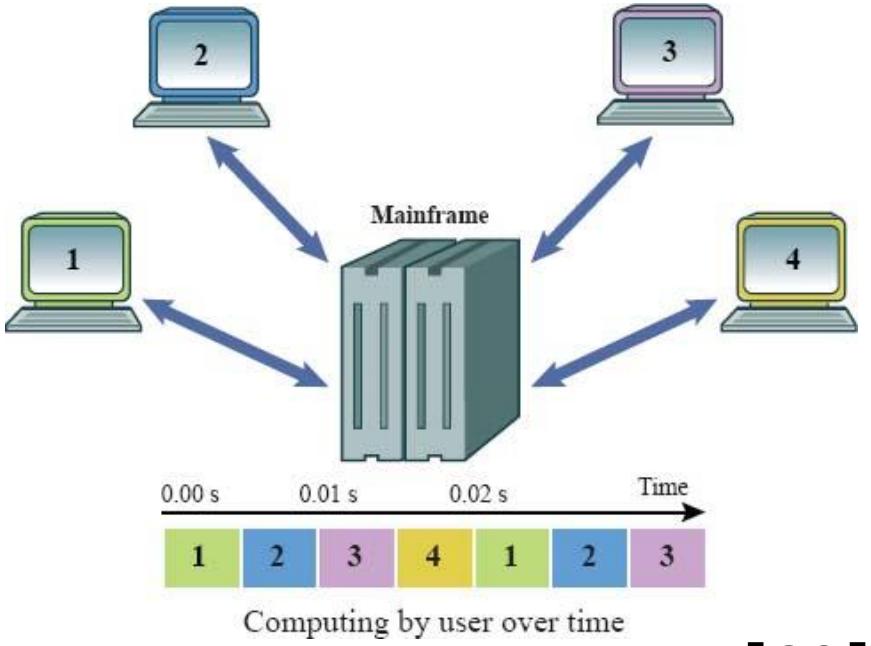

*Gravt förenklad

# Time-Sharing Systems

## Multi-Tasking

## Multi-User



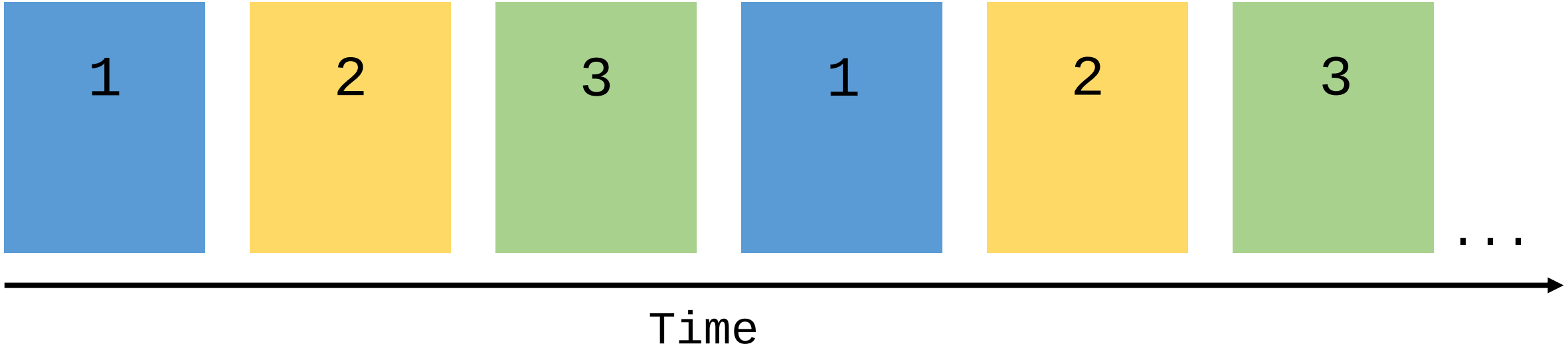

[28]

# Time-Sharing Systems

## Multi-Tasking

Flera program kan köras ’parallellt’

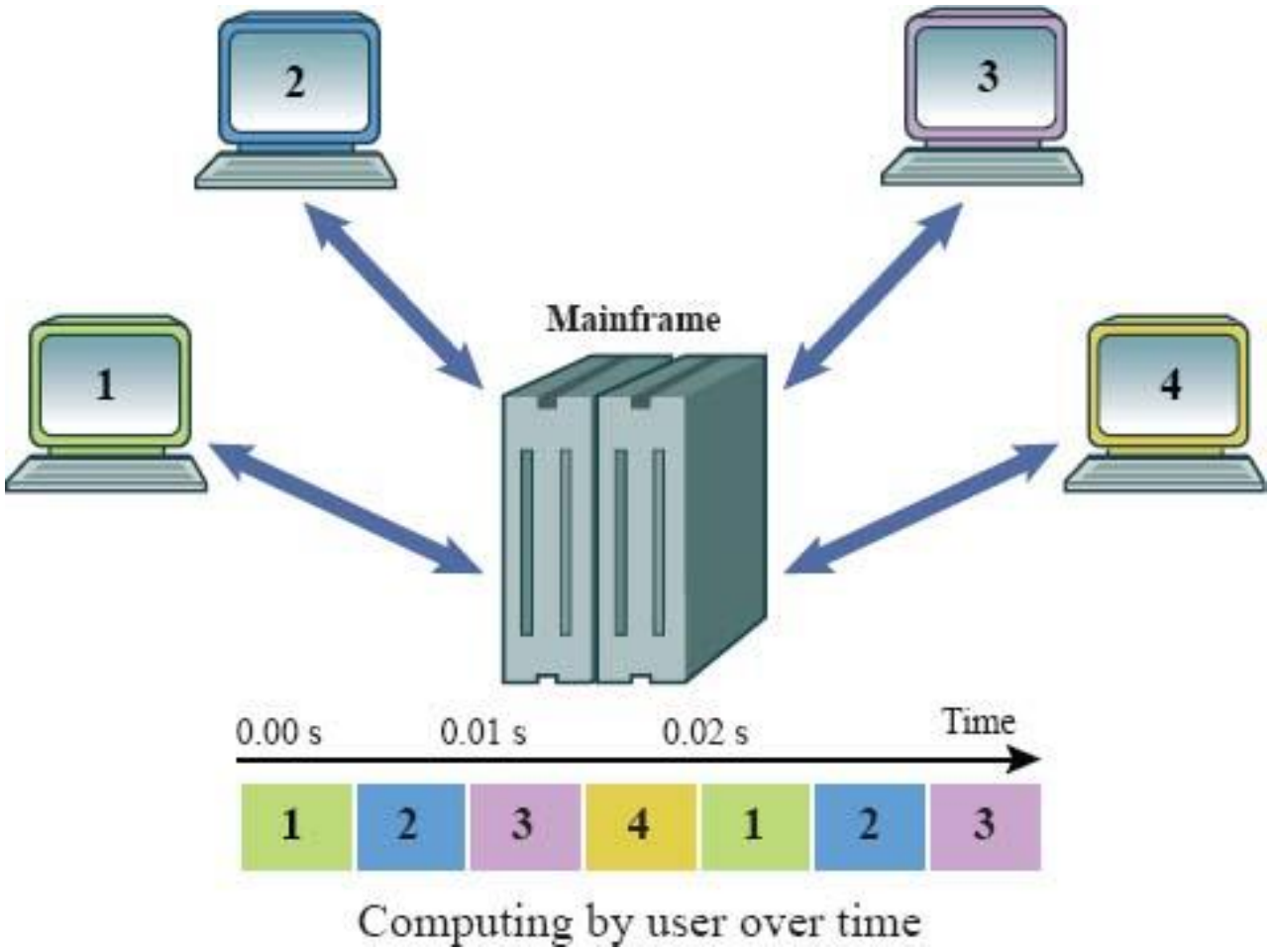
En i taget, men växlar snabbt mellan dom



# Time-Sharing Systems

## Multi-User

- Flera användare
- Växlar mellan dom



# Time-Sharing Systems

Flera program/användare delar på samma resurser

→ Större krav på säkerhet och resurshantering

# 3. Säkerhet

# Operativsystemssäkerhet

Operativsystemet styr över tillgång till hårdvara och ser till att:

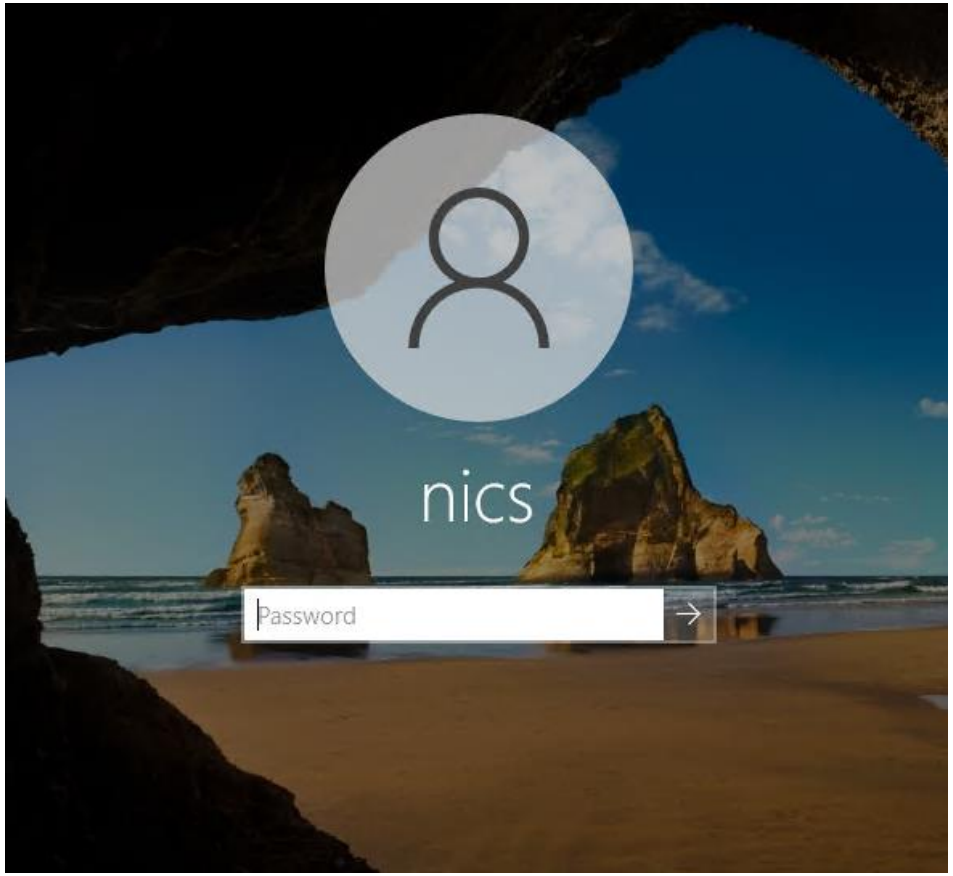
- Den förblir i kontroll
- Alla användare/program får lika tillgång till hårdvara
- Ingen kan roffa åt sig resurser på andras bekostnad

# Operativsystemssäkerhet

Operativsystemet hanterar autentisering av användare

’Är du den du utger dig för att vara?’

→ Lösenord

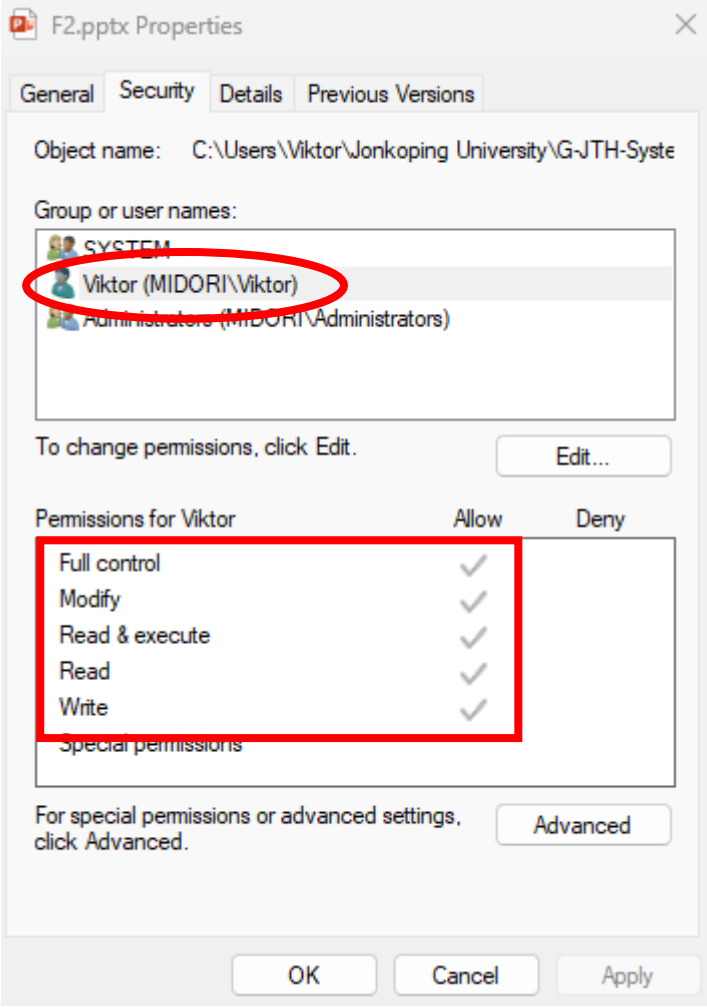


# Operativsystemssäkerhet

Operativsystemet begränsar vad användare & processer får göra

Rättigheter/behörigheter (eng. permissions)

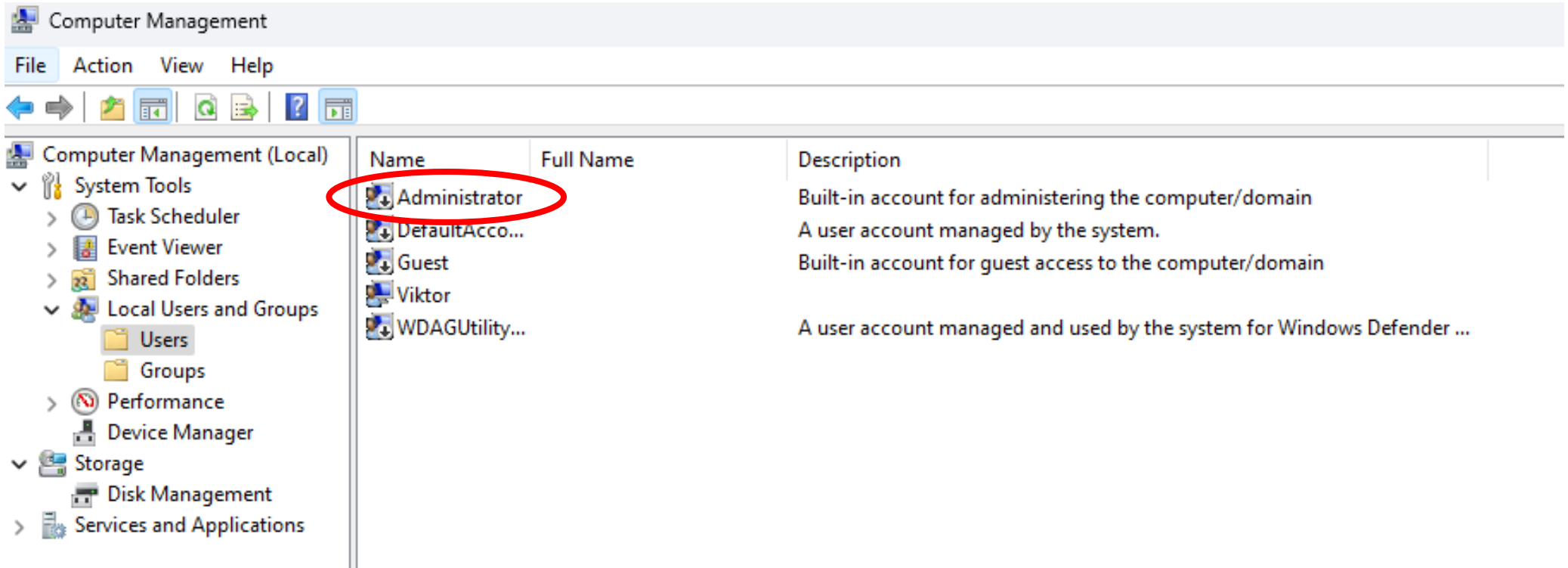
Skyddar på så sätt användare från varandra



# Operativsystemssäkerhet

Det behöver finnas en användare för att utföra känsligare uppgifter

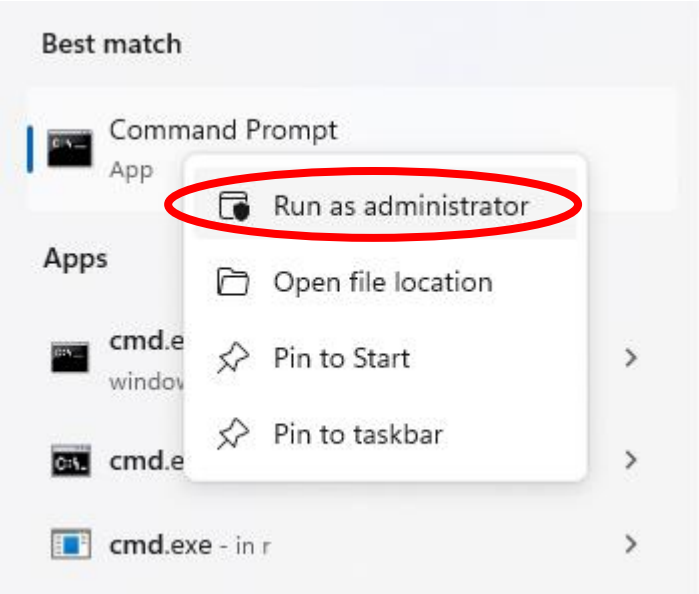
Windows: *Administrator*

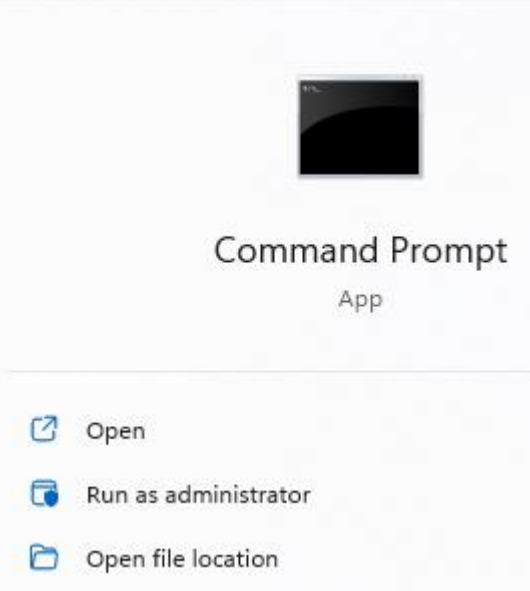


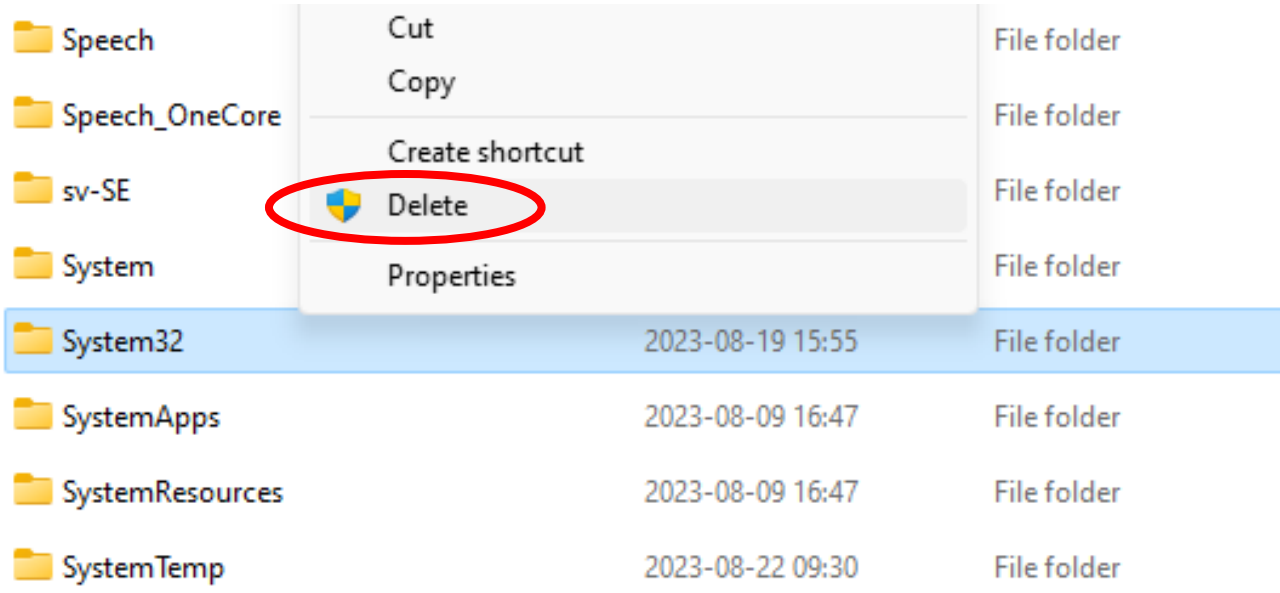
# Operativsystemssäkerhet

Kan utföra saker som en annan användare

T.ex. *Run as Administrator*



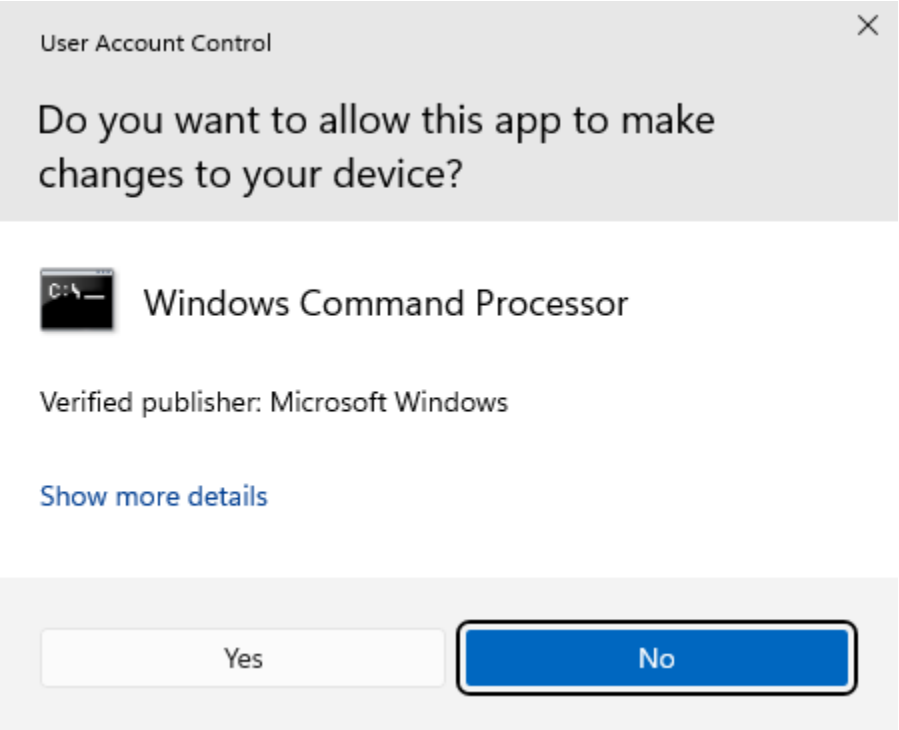



# User Account Control

I Windows används **UAC** för att utföra saker som en annan användare

Default: Adminstrator

Går även att lösenordsskyddad



# Vilka rättigheter har processer?

UNIX-liknande OS:

- ’Ärver användarens rättigheter’
- Körs med samma rättigheter som den som startar programmet

# Vilka rättigheter har processer?

Windows:

- Körs med begränsade standard rättigheter
- Men kan be om utökade rättigheter genom **UAC**

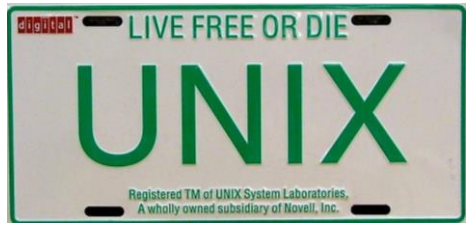
# 4. Historia

# Operativsystems historia

Två ’familjeträd’

- *UNIX*
- *Microsoft*

# Tidslinje över UNIX-liknande OS*




[1]

1969

Ett tidigt **time-sharing** system

Påbörjades av *Ken Thompsson* & *Dennis Ritchie* på *Bell Labs*

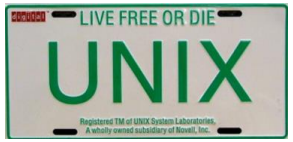
[2]

*förenklad & inte skalenlig

# Tidslinje över UNIX-liknande OS*

BSD

1969



[1]

1978

[3]

Original *UNIX* används inte längre

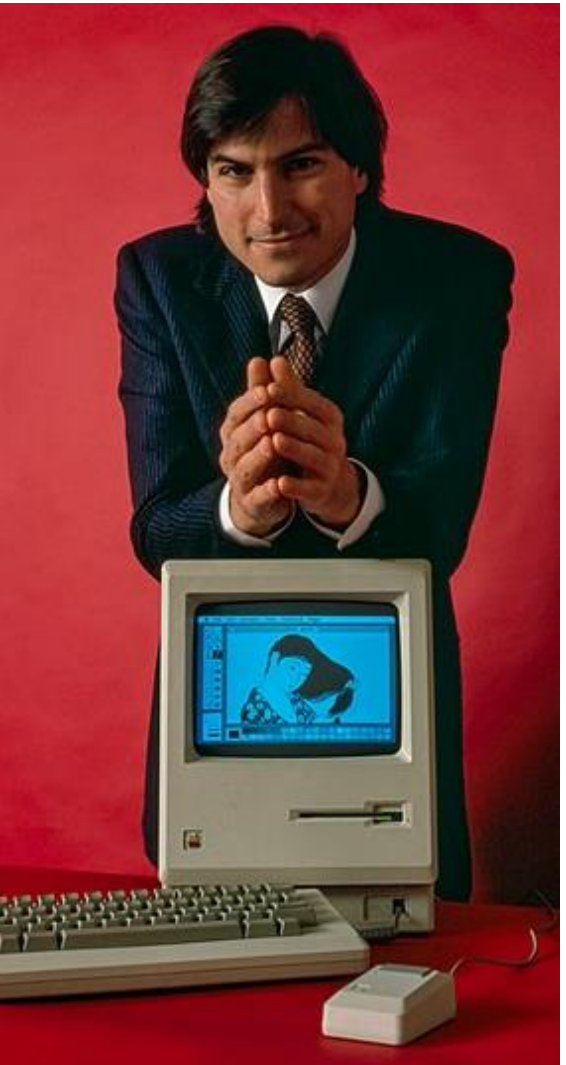
Varianter av *BSD*, ifrån Berkely University, är dock fortfarande populära

[4]

*förenklad & inte skalenlig

# Tidslinje över UNIX-liknande OS*

## Apple



[6]

[5]

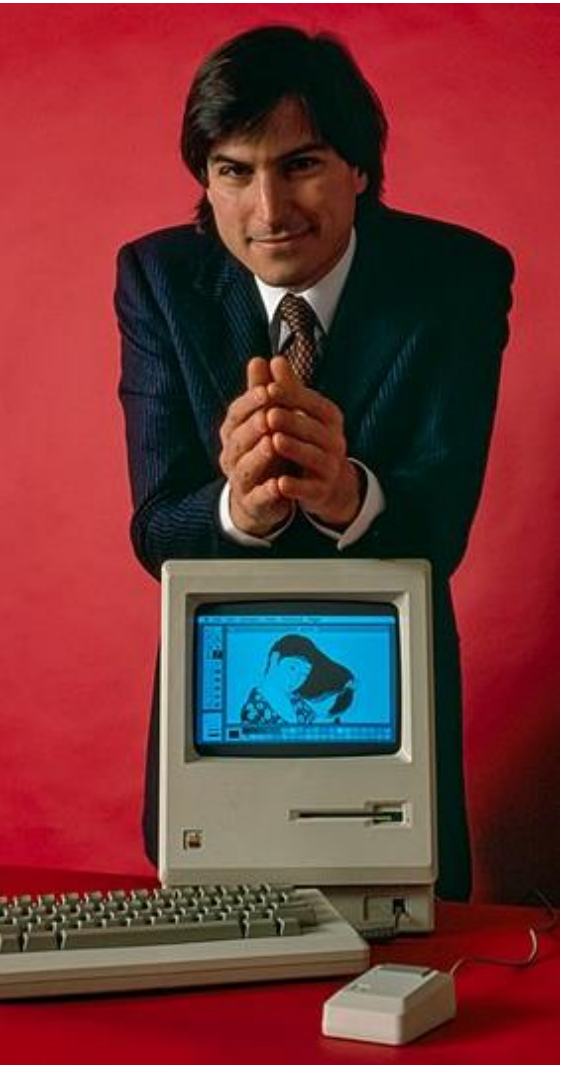
1984

*Macintosh*, en tidig GUI-baserad persondator

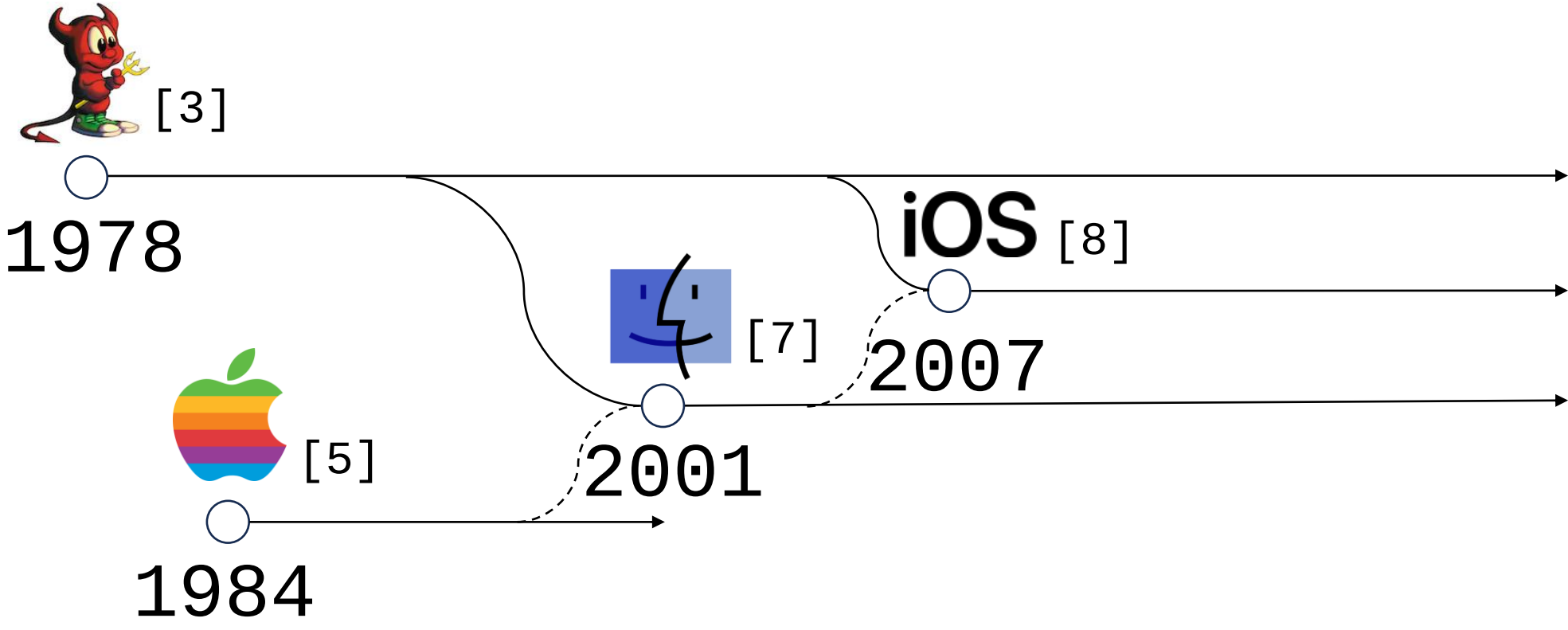
*förenklad & inte skalenlig

# Tidslinje över UNIX-liknande OS*

## Apple



[6]

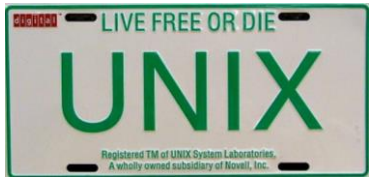



Blev senare *macOS* och *iOS*

*förenklad & inte skalenlig

# Tidslinje över UNIX-liknande OS*

Linux




[1]

[9]

1969

1992

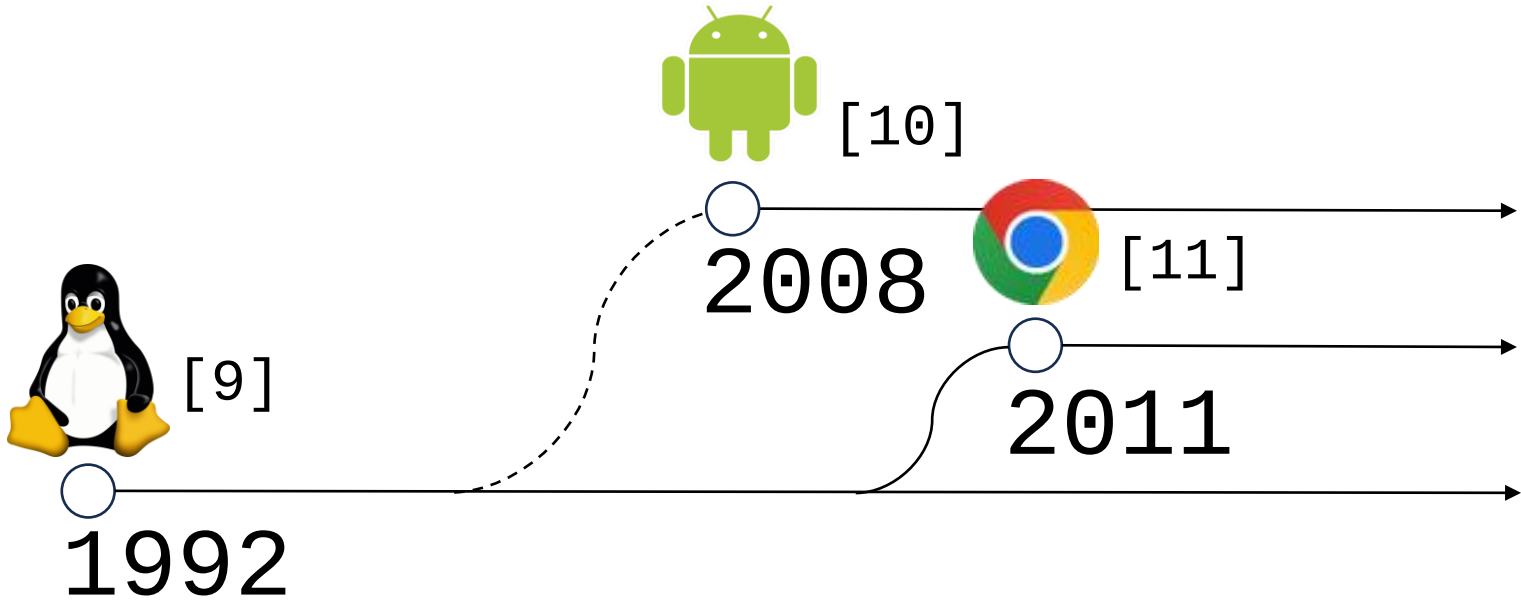
*Linux*, antagligen det vanligaste operativsystemet

Mer i *Systemadministration – Linux*

*förenklad & inte skalenlig

# Tidslinje över UNIX-liknande OS*

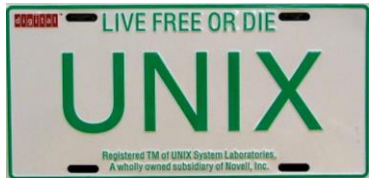
## Google




Baserat på Linux har Google byggt både *Android* och *ChromeOS*

*förenklad & inte skalenlig

# Tidslinje över UNIX-liknande OS*




[1]

1969

[3]

1978

[5]

1984

[9]

1992

[7]

2001

iOS [8]

2007

[10]

2008

[11]

2011

*gravt förenklad & inte skalenlig

# Tidslinje över Microsoft OS*

## MS-DOS

[12]

1981

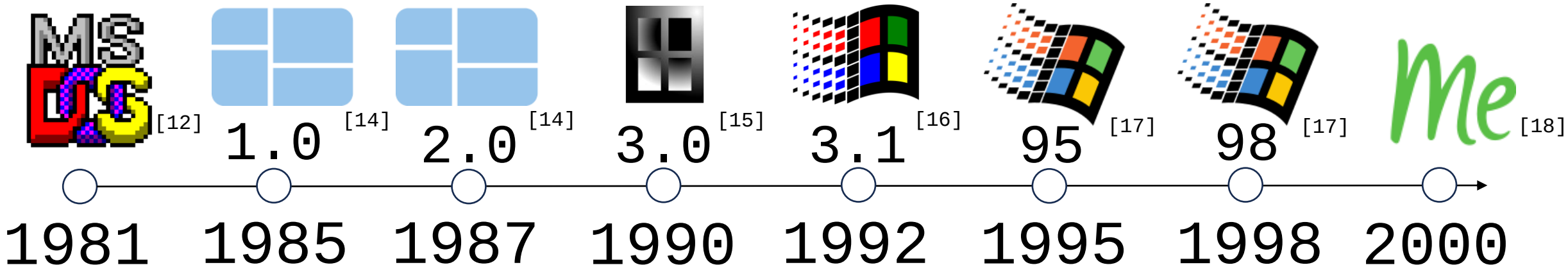
Microsoft utvecklade ett CLI-baserat OS åt IBM datorn *Personal Computer ‘PC’*

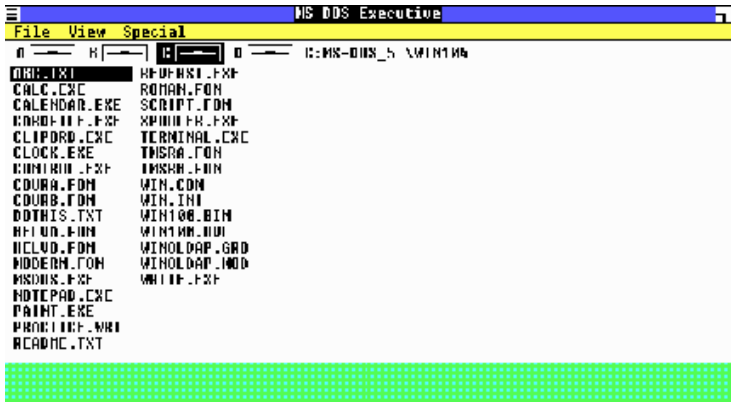
[13]

*förenklad & inte skalenlig

# Tidslinje över Microsoft OS*

## MS-DOS




[19]

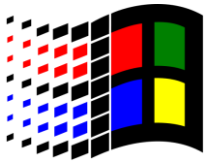
Baserat på MS-DOS utvecklades *Windows*

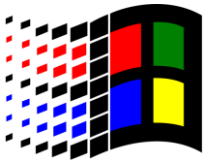
En grafisk *desktop* miljö som sakta ersatte MS-DOS CLI

*förenklad & inte skalenlig

# Tidslinje över Microsoft OS*

NT


[16]


[16]

[17]

[20]

NT 3.1 — 1993

NT 3.5 — 1994

NT 4.0 — 1996

2000 — 1999

Parallellt med DOS-baserade *Windows* utvecklades även en ny *Windows* familj

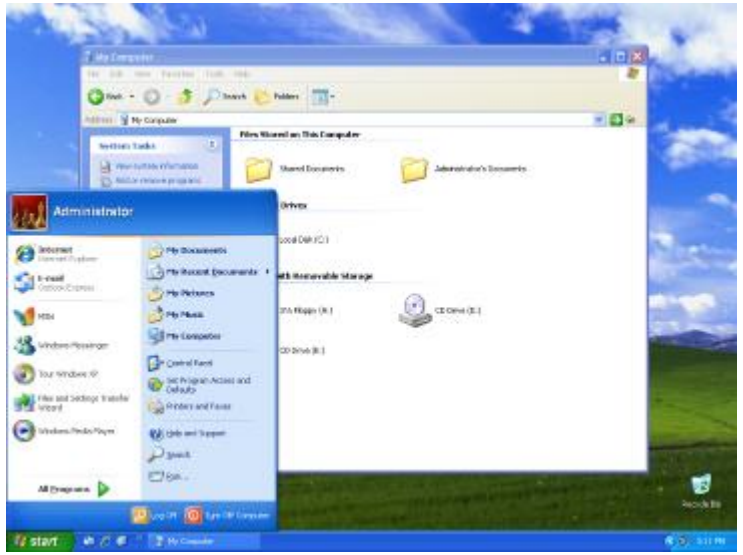
[21]

*förenklad & inte skalenlig

# Tidslinje över Microsoft OS*

NT

| XP | Vista | 7 | 8 | 8.1 | 10 | 11 |
|---|---|---|---|---|---|---|
| [21] | [22] | [22] | [23] | | [24] | [25] |
| 2001 | 2007 | 2009 | 2012 | 2012 | 2015 | 2021 |



…vilket alla moderna versioner av *Windows* är baserade på [27]

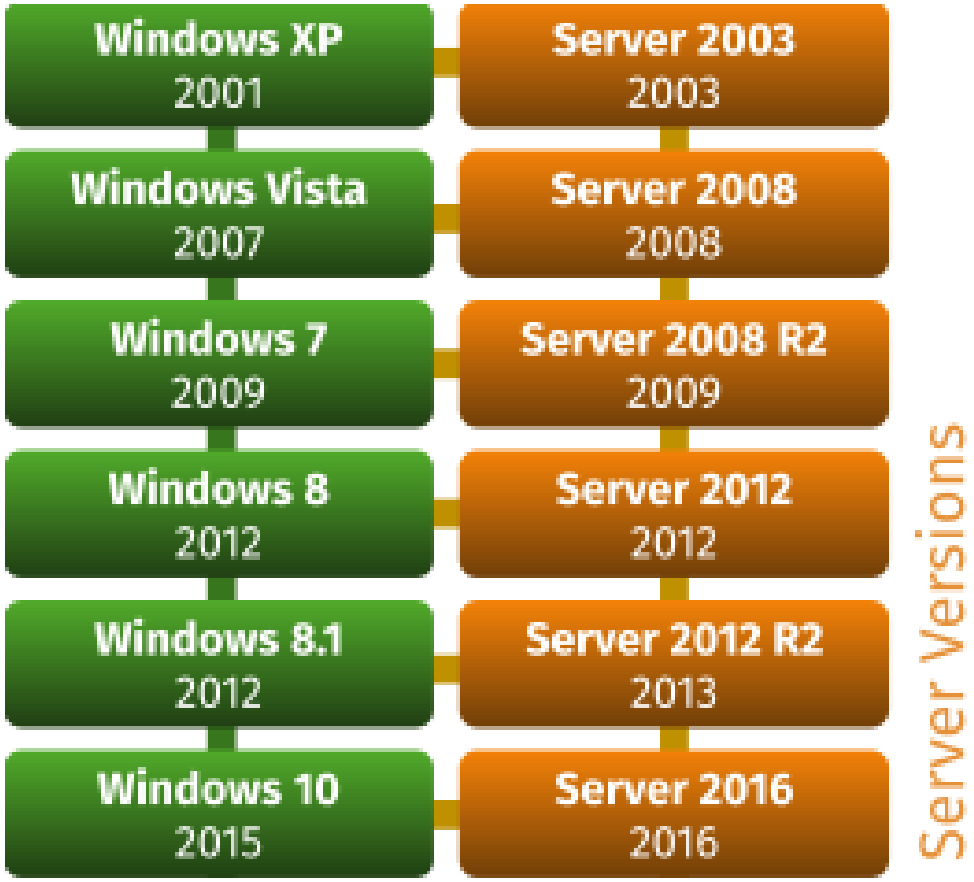
*förenklad & inte skalenlig

# Tidslinje över Microsoft OS

## Serveroperativsystem

Microsoft har även en anpassad version av *Windows* för servrar

Senaste versionen släpptes 2021 och är baserad på *Windows 11*




*A flowchart which represents the version history of Windows*, by CrustyCaverns, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=128808419). CC0 1.0

# Till nästa gång

- Läs kurslitteraturen
- Se video-demonstrationer
- Fredag: laboration

Tack för idag

# Bildreferenser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